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Reliability estimation for individual predictions in machine learning systems: A model reliability-based approach

Xiaoge Zhang [a,*], Indranil Bose [b]

[a] Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, Kowloon, Hong Kong
[b] Department of Information Systems, Supply Chain Management & Decision Support, NEOMA Business School, 59 rue Pierre Taittinger, Reims 51100, France

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ABSTRACT

The conventional aggregated performance measure (i.e., mean squared error) with respect to the whole dataset would not provide desired safety and quality assurance for each individual prediction made by a machine learning model in risk-sensitive regression problems. In this paper, we propose an informative indicator $\mathcal{R}(\mathbf{x})$ to quantify model reliability for individual prediction (MRIP) for the purpose of safeguarding the usage of machine learning (ML) models in mission-critical applications. Specifically, we define the reliability of a ML model with respect to its prediction on each individual input $\mathbf{x}$ as the probability of the observed difference between the prediction of ML model and the actual observation falling within a small interval when the input $\mathbf{x}$ varies within a small range subject to a preset distance constraint, namely $\mathcal{R}(\mathbf{x}) = P(|y^* - \hat{y}^*| \leq \varepsilon | \mathbf{x}^* \in B(\mathbf{x}))$, where y^* denotes the observed target value for the input $\mathbf{x}^*$, $\hat{y}^*$ denotes the model prediction for the input $\mathbf{x}^*$, and $\mathbf{x}^*$ is an input in the neighborhood of $\mathbf{x}$ subject to the constraint $B(\mathbf{x}) = \{\mathbf{x}^* | \ \|\mathbf{x}^* - \mathbf{x}\| \leq \delta\}$. The developed MRIP indicator $\mathcal{R}(\mathbf{x})$ provides a direct, objective, quantitative, and general-purpose measure of "reliability" or the probability of success of the ML model for each individual prediction by fully exploiting the local information associated with the input $\mathbf{x}$ and ML model. Next, to mitigate the intensive computational effort involved in MRIP estimation, we develop a two-stage ML-based framework to directly learn the relationship between $\mathbf{x}$ and its MRIP $\mathcal{R}(\mathbf{x})$, thus enabling to provide the reliability estimate $\mathcal{R}(\mathbf{x})$ for any unseen input instantly. Thirdly, we propose an information gain-based approach to help determine a threshold value pertaing to $\mathcal{R}(\mathbf{x})$ in support of decision makings on when to accept or abstain from counting on the ML model prediction. Comprehensive computational experiments and quantitative comparisons with existing methods on a broad range of real-world datasets reveal that the developed ML-based framework for MRIP estimation shows a robust performance in improving the reliability estimate of individual prediction, and the MRIP indicator $\mathcal{R}(\mathbf{x})$ thus provides an essential layer of safety net when adopting ML models in risk-sensitive environments.

1. Introduction

Artificial intelligence (AI), particularly machine learning (ML) algorithms,[1] have been developing at a fast pace over the past decades to tackle a broad spectrum of regression problems [1–4], such as disease severity prediction, aviation safety assessment, and financial risk analysis, to name a few. While ML shows enormous business values and customer benefits, it is also giving rise to a series of unintended, and in some cases serious consequences. In fact, the absence of safety assurance component in ML-based solutions for regression problems has severely hindered its adoption in life and safety-critical applications [5,6], such as drug industry, aerospace system, and medical diagnosis. Currently, the use of ML in high-stakes decision settings is still in its infancy compared with other areas (e.g., natural language processing, object recognition) because life and safety-critical applications are not only concerned with having a properly trained ML model, but they are more concerned with the understanding of scenarios and mechanisms when ML model malfunctions or even fails. Failing to identify the likely sources of error in risk-sensitive applications could lead to disastrous outcomes (i.e., huge economic loss) and in some cases might put human life in danger. For example, in a recent road test conducted at the Maricopa County, Arizona, U.S.A., the automated system in an Uber self-

* Corresponding author.
E-mail address: xiaoge.zhang@polyu.edu.hk (X. Zhang).

[1] We use the abbreviation AI and ML to refer to "artificial intelligence" and "machine learning" in the remainder of the paper.

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driving car failed to recognize a pedestrian crossing midblock, which led to a mishap of the pedestrian fatality [7]. The National Transportation Safety Board (NTSB) investigation concluded that Uber's "inadequate safety risk assessment procedures" and "ineffective oversight of vehicle operators" were the primary contributing factors [8].

In addition, notwithstanding the revolutionary impact of ML systems, the translation rate of ML algorithms into practical solutions for high-stakes decision makings is relatively low [9] because the field of ML fundamentally lacks a systematic framework for reasoning about risk, uncertainty, and their potential catastrophic outcomes, while safety and quality implications are of central importance in critical applications. Nowadays, a common practice when evaluating the quality of a ML model in the context of regression tasks is to employ some aggregated performance metrics [10–13], such as mean squared error (MSE) and mean absolute error (MAE) on some test dataset. These aggregated measures are useful for the general understandings of the statistics of model performance at an aggregated level, and it is sufficient to count on them to deploy ML models in risk-free applications. However, in risk-sensitive environments, such as medical diagnosis and manufacturing industries, it is risky to apply a trained ML model by merely relying on these aggregated performance metrics. A general performance measure with respect to the whole dataset will not provide desired safety and quality warranty for each individual input in high-stakes environments [14–16]. In this regard, blindly accepting ML model predictions might lead to significant consequences for people and the broader society. For example, Nushi et al. [17] showed that a ML model with a small MSE can still make large pointwise errors on individual predictions for certain types of inputs. Besides, the aggregated performance measures have limited expressiveness in characterizing the magnitude of error or level of uncertainty with respect to each specific individual prediction, while such information is essential for risk assessment, risk management as well as customized decision making in high-stakes circumstances. Take the severity estimation of Parkinson based on the Parkinson's Disease Rating Scale as an example [18], a ML model might generate the same score (i.e., 112) for two different patients, that is, the two patients have the same level of disease severity as predicted by the ML model. However, the reliability of model prediction could vary significantly for the two individual patients due to a variety of reasons [19], for example, because the two patients belong to different age groups, or they have different medical history, or different amounts of data for each age group is available for ML model training. If we take model reliability into consideration, it is possible that a younger patient might have its score as 112 ± 30, while the same prediction for an older patient might be much more reliable (i.e., 112 ± 5). The apparent difference in model reliability as reflected by the standard deviation of ML model prediction has a non-negligible impact on the subsequent decision making pertaining to the plan, design, and implementation of individualized treatment for each patient.

Unfortunately, the issue of model reliability for individual prediction (MRIP)[2] in the context of machine learning has not been investigated as comprehensively as needed. Particularly, we focus on developing a quantitative metric to estimate the reliability of ML prediction for regression problems in this study. It should be noted that the past research has referred to MRIP with different terms in ML [20,21], such as prediction confidence interval, prediction uncertainty, prediction error estimate. Towards MRIP estimation, several studies have developed multiple quantitative metrics to measure the reliability of pointwise ML prediction in classification problems from a variety of perspectives. For example, Nguyen and Hullermeier [22] introduced an uncertainty-driven abstention mechanism into the ML workflow in a multi-label classification problem so that the model learned to partially abstain from making predictions and predicted only parts of it, namely those for which it was certain enough. Bosnić and Kononenko [20] provided an overview on the methods developed for MRIP estimation, and the methods were grouped into three classes: perturbation-based methods, sensitivity analysis-based approaches, and unlabeled data-based methods.

Even though some progress has been made along the development of approaches for MRIP estimation in classification problems, the issue of MRIP for numerical prediction models has been under-explored while it undoubtedly represents an important issue that deserves increasing attention due to the broad implications of MRIP in safety-critical applications. Along this front, Schulam and Saria [23] proposed a resampling uncertainty estimation algorithm to audit pointwise reliability of model predictions. Adomavicius and Wang [19] developed a general-purpose ML-based framework to estimate ML model reliability for individual prediction, where the estimated absolute prediction error between the ML model prediction and the actual value was used as a quantitative indicator of reliability. In spite of the progress, the development of MRIP is still in the early stage and several crucial issues remain unresolved. ML systems, similar to other types of computational model predictions, face aleatory and epistemic uncertainty arising from diverse sources [24], such as measurement noise, missing data, lack of data, model structure and parameter uncertainty. These uncertainties, both individually and collectively, have a significant impact on the accuracy of ML model prediction. Development of approaches for accurate MRIP modeling in consideration of these different sources of uncertainties will safeguard the usage and enhance the adoption of ML models in safety and quality-critical applications. From this standpoint, it is of paramount importance to provide an accurate estimation of MRIP to indicate the level of confidence with which we accept or reject the model prediction. In this regard, the knowledge of MRIP provides valuable insights on model predictive performance specific to each input, and such knowledge can be critical in risk-sensitive fields. Successful identification of model predictions that are deemed unreliable beforehand will help prevent ML system from causing unintended outcomes or harmful behaviour in practice.

In this paper, we are motivated to develop a novel MRIP indicator to assess the reliability of ML models. Specifically, we define a MRIP indicator $\mathcal{R}(\mathbf{x})$ to measure the reliability of the ML model with respect to its prediction on the input $\mathbf{x}$ as the probability of the observed difference between the model prediction and the actual value falling within a small interval when the input $\mathbf{x}$ varies within a predefined range. The proposed MRIP metric is devised to exploit two fundamental principles when judging whether a ML model prediction is reliable: the density criteria and the local fit criteria [25]. The density criteria states that a prediction at input $\mathbf{x}$ is reliable if there are samples in the training data that are similar to $\mathbf{x}$; the local fit criteria states that a prediction for input $\mathbf{x}$ is reliable if the ML model has a small error on samples that are in the vicinity of the input $\mathbf{x}$. With the reliability indicator $\mathcal{R}(\mathbf{x})$, we develop an approach to approximate its value by taking advantage of the information of neighboring points around $\mathbf{x}$. Next, to mitigate the intensive computational effort, we formulate a two-stage ML-based framework to estimate the value $\mathcal{R}(\mathbf{x})$ for unseen inputs. Compared to the state-of-the-art literature, the main contributions of this paper are summarized as follows:

1. We develop a general-purpose reliability indicator $\mathcal{R}(\mathbf{x})$ to quantify the MRIP of ML model. The proposed reliability estimator $\mathcal{R}(\mathbf{x})$ has the advantage of modeling different types (either individually or collectively) of heterogeneous noise (e.g., measurement noise in input, algorithmic bias, inherent variability in the underlying system to collect input data) arising from the input $\mathbf{x}$ and the ML model $\widehat{y} = \widehat{f}(\mathbf{x})$ as demonstrated in Section 4.3. The computational results in Section 4 suggest that the proposed MRIP indicator $\mathcal{R}(\mathbf{x})$ provides an essential layer of safety net when deploying ML models in practice as it provides extra information on the potential error of individual

[2] In the rest of the paper, we use the abbreviation MRIP to refer to "model reliability for individual prediction"

predictions, thus preventing the ML model from causing any unintended outcomes.

2. We rigorously examine the relationship between $\mathcal{R}(\mathbf{x})$ and the magnitude of the underlying noise. To this end, a sampling-based approach is developed to derive the MRIP value provided the underlying process of noise generation is known. Next, we develop a practical approach to approximate MRIP value $\widetilde{\mathcal{R}}(\mathbf{x})$ using the neighboring points around $\mathbf{x}$ (see Section 3.1). The high correlation between the MRIP values derived by the sampling approach and estimated by the approximation method indicates that estimating the MRIP value by neighboring points provides a reasonable approximation to the actual MRIP value.
3. We conduct comprehensive computational experiments to compare the performance of the proposed two-stage ML-based framework for MRIP estimation with other state-of-the-art reliability estimates. Towards this end, we train nine ML models with eight commonly used datasets from the UCI machine learning repository. The correlation-based performance assessment suggests that there is a good agreement between $|y-\widehat{y}|$ and the ML-based estimation of MRIP value $\widehat{\mathcal{R}}(\mathbf{x})$, which outperforms other alternatives across all the datasets. In addition, the developed approach has a desirable feature: the larger the ϵ (or the larger the risk), the higher the proportion of instances violating the prediction accuracy requirement (e.g., $|y-\widehat{y}|>\epsilon$).

The rest of this paper is structured as follows. In Section 2, we review the state-of-the-art approaches for MRIP estimation and assessment. In Section 3, we introduce the two-stage ML-based framework for MRIP estimation. In Section 4, we demonstrate the procedures of the proposed methodology, and present the computational results using both synthetic and real-world datasets. In Section 5, we discuss the implications of the developed MRIP indicator for practical adoptions of ML models. In Section 6, we conclude this study and discuss future research directions.

2. Literature review

Considering the increasing reliance on ML algorithms from a broad range of applications, it is getting abundantly clear that having a comprehensive and nuanced comprehension of ML model performance play a cornerstone role in assuring the fairness and reliability of ML systems. Towards this end, there is a pressing demand for MRIP modeling and estimation. Accurate MRIP modeling and estimation has significant implications on the interpretations, operations, maintenance, and management of ML algorithms in practice [26,27]. In particular, precise MRIP estimation acts as an essential layer of safety net when adopting ML models in high-stakes decisions as MRIP facilitates end-users to have a better understanding for which data points the ML model is expected to have good or poor performance, thus making risk-informed decisions accordingly. Regarding MRIP modeling and estimation, the state-of-the-art approaches can be roughly grouped into three broad categories:

1. Transduction-based approaches. A transductive reliability framework typically consists of an inductive step and a transductive step. The difference of ML prediction between the inductive and transductive step is then used to estimate the reliability with respect to the ML prediction for the individual input $\mathbf{x}$.
2. Distribution-based methods. That is to build a distribution as the prediction of the ML model with respect to an individual input $\mathbf{x}$ via a variety of models, such as Bayesian machine learning [28,29], Gaussian process [30,31].
3. Indicator-based methods. This line of methods aims to provide a single numerical value to reflect model reliability with respect to the individual input $\mathbf{x}$. Some common measures in this group of methods include model prediction error-based indicator [19], prediction interval-based indicator [32,33].

Transduction-based methods are rooted in transductive reasoning. Transductive reasoning is an inference principle that reasons from case to case, and it features an inductive step followed by a transductive step when estimating element-wise ML model reliability. In the inductive step, a ML model is trained on an initial training dataset and induces a prediction y_{ind} on an individual input $\mathbf{x}$. Next, in the transductive step, the ML model is retrained with an extended training dataset by adding the individual example $\mathbf{x}$ and transduces an updated prediction y_{trans} for the same input $\mathbf{x}$. The difference between the prediction of inductive y_{ind} and transductive classifiers y_{trans} is then used to build a quantitative reliability measure to indicate the ML model prediction performance on the individual input $\mathbf{x}$. Transduction-based methods have some clear advantages as they can be used with any kind of learning algorithm. For example, Bosnić and Kononenko [20,34] extended the idea of transductive reasoning to assess the reliability of model prediction in regression problems. Nouretdinov et al. [35] investigated the application of transductive reasoning in functional and structural MRI data to generate diagnostic and prognostic decisions at the individual level. Despite the simplicity and easiness to implement, transduction-based methods suffer from poor interpretability to justify the derived reliability indicator. In particular, it is intricate for end users to understand the subtle difference between the ML model derived at the inductive and transductive step due to the high complexity, while it is a necessity to provide informative explanations on reliability indicators in high-stakes decision-making environments. Bosnić and Kononenko [36] developed an approach based on the sensitivity analysis of the output of regression models with respect to small changes in the input dataset to estimate individual prediction reliability.

Towards reliability estimation at the individual level, another approach prevailing in the literature is to generate a distribution as ML model prediction by taking a Bayesian approach. Typically, the dispersion of the predictive distribution in the context of regression is used as a quantitative indicator of MRIP. In general, model prediction with wider width for a given confidence interval indicates a lower MRIP. In Bayesian machine learning, it is a prerequisite to assume some prior distributions (e.g., distribution form, distribution parameters) to characterize the parameters of ML models, the assumed prior distributions are then combined with the likelihood function to infer the posterior distributions of ML model parameters via maximum likelihood estimation, maximum a posterior [37], or variational inference [38,39]. Afterwards, random samples are generated from the posterior distributions of ML parameters to generate the predictive distribution for any unseen input. However, Bayesian approach is model-specific as it usually requires sophisticated procedures to develop the Bayesian version of a given ML model [40]. In other words, the procedures of transforming a deterministic ML model into its corresponding Bayesian counterpart vary from model to model and from assumption to assumption. The derivations of the Bayesian ML model closely depend on the specific statistical assumptions. In addition, certain statistical assumptions in the Bayesian context might get violated in practice. For example, the homoscedasticity assumption in the case of Bayesian linear regression might get violated in practice as the underlying noise in the observed data might vary with the specific location of the individual input $\mathbf{x}$ [41]. In the field of Bayesian methods, Gaussian process [42] is another popular method commonly adopted in the literature. The advantage of Gaussian process over other Bayesian ML models is that Gaussian process has a rigorous mathematical foundation, and it enables to estimate model prediction uncertainty in a closed form, which significantly facilitates the derivation of confidence interval. However, in Gaussian process, the prediction variance on a new observation $\mathbf{x}^*$ only depends on the relative location of $\mathbf{x}^*$ with respect to other observations in the training data *in the feature space*, and it has nothing to do with the values

of the observed target [19]. Due to this deficiency, Gaussian process is unable to capture the magnitude of prediction error arising from the variability pertaining to the observed target. As a consequence, the inability of Bayesian approaches failing to characterize the heteroscedasticity pertaining to the noise in the observed target eventually propagates and results in a misleading MRIP estimation.

Indicator-based approaches typically generate a single value as a measure of model prediction reliability for any individual input $\mathbf{x}$. For example, VarBag [43], VarA and MSE [44], and VarP [45] are some representative indicator-based methods to estimate the reliability of individual ML prediction, see more details on their definition and calculation in Section 4. Indicator-based approaches have two explicit advantages over Bayesian methods: (I) there is no need to make any statistical assumptions in the indicator-based methods; (II) indicator-based methods are model-agnostic. In other words, they can be combined with any ML model in a straightforward manner. Among these approaches, early studies concentrate on developing ensemble methods to estimate the prediction variance or prediction interval specific to an individual input. For example, Lakshminarayanan et al. [46] developed a simple and scalable non-Bayesian method to estimate predictive uncertainty of neural network by building an ensemble of neural networks. Another prevailing way to build the MRIP indicator is to exploit the local information of the individual input $\mathbf{x}$ to approximate the variability of underlying data. For example, Bosnić and Kononenko [45] developed a density-based approach to identify neighbors around the individual input $\mathbf{x}$ and derived the reliability estimate of $\mathbf{x}$ using the nearest neighbors' information. Intuitively, the density-based approach is rooted in the logic that error is lower for predictions which are made in the denser problem subspaces and higher for predictions which are made in the sparser subspaces. Similarly, density-based methods might not be helpful in the heteroscedastic settings because of their failure in modeling the variability pertaining to the observed target.

3. A ML-based framework for MRIP modeling and estimation

In this section, we describe the proposed ML-based framework for MRIP modeling and estimation. The proposed framework consists of three steps. In the first step, we define a general-purpose indicator $\mathcal{R}(\mathbf{x})$ to model MRIP, and the proposed indicator $\mathcal{R}(\mathbf{x})$ accounts for the variability associated with the input $\mathbf{x}$ and the ML model $\widehat{y} = \widehat{f}(\mathbf{x})$ simultaneously. Next, we develop two different approaches to derive the value of MRIP: sampling-based approach and neighboring points-based approximation. Afterwards, we formulate a two-stage ML-based methodology to estimate MRIP for the sake of mitigating the computational effort involved in the neighboring points-based approximation approach. Thirdly, an information gain-driven method is used to determine the threshold value of $\mathcal{R}(\mathbf{x})$ so as to decide when to abstain from depending on the ML model prediction.

3.1. Model reliability for individual prediction Indicator: An introduction

3.1.1. MRIP modeling

In general, various sources of uncertainties can be roughly grouped into two classes:

1. Aleatory uncertainty: It usually stems from natural variability and is irreducible in nature. Examples of aleatory uncertainty include measurement noise (e.g., sensor noise), natural variability (e.g., variability in temperature).
2. Epistemic uncertainty: Epistemic uncertainty refers to the uncertainty of the model due to things one could in principle know, but does not in practice. In the context of ML, epistemic uncertainty often arises because the ML model neglects certain effects or overlooks certain important variables, the amount of training data is scarce, or the values of some variables are missing. Epistemic uncertainty typically gets manifested in the form of model parameter (or structure) uncertainty and model form uncertainty (e.g., which ML model to use).

These diverse sources of uncertainties eventually get manifested as discrepancy or variation associated with the input or the output of the ML model. Considering the uncertainties pertaining to the model input (e.g., measurement noise) as well as the ML model (e.g., uncertainty of model structure and model parameters, model bias), we treat the difference between the actual observation and the ML model prediction ($|y - \widehat{y}|$) as an uncertain quantity. Pursuing along this path, we define a simple and intuitive indicator to measure the ML model reliability, that is the probability of the observed difference $|y - \widehat{y}|$ falling within a small interval.

In particular, considering a ML model $\widehat{y} = \widehat{f}(\mathbf{x}, \boldsymbol{\theta})$, where $\boldsymbol{\theta}$ denotes the parameters of the ML model, we define the reliability of the ML model with respect to its prediction for the individual input $\mathbf{x}$ as follows:

$$\begin{aligned} &\mathcal{R}(\mathbf{x}) = P(|y^* - \widehat{y}^*| \le \varepsilon \,|\mathbf{x}^* \in B(\mathbf{x})\,), \\ &\text{s.t.} \quad \mathrm{B}(\mathbf{x}) = \{\mathbf{x}^* |\ \ \|\mathbf{x}^* - \mathbf{x}\| \le \delta\}, \end{aligned} \tag{1}$$

where y^* and $\widehat{y}^*$ represent the observed target value and the ML model prediction for the individual input $\mathbf{x}^*$, and $\mathbf{x}^*$ is an input in the neighborhood of $\mathbf{x}$. $B(\mathbf{x})$ denotes the set of neighboring points $\mathbf{x}^*$ around the input $\mathbf{x}$ subject to the constraint $\|\mathbf{x}^* - \mathbf{x}\| \le \delta$.

δ and ε are two non-negative parameters specifying the accuracy requirement with respect to the neighboring points of the input $\mathbf{x}$ and model output $\widehat{y}^*$, respectively. The values of δ and ε can be configured according to the variance of y and the number of features in $\mathbf{x}$. In general, the higher the variance of y and the more the number of features in $\mathbf{x}$, the higher the value of ε and the larger the radius δ; and vice versa. $\|\mathbf{x}\|$ denotes the $\mathscr{L}2$ norm of $\mathbf{x}$ [47], and $|y^* - \widehat{y}^*|$ refers to the absolute value of $y^* - \widehat{y}^*$. In Eq. (1), the two parameters δ and ε control the accuracy requirement of the MRIP, and they help to estimate the model reliability $\mathcal{R}(\mathbf{x})$. The smaller δ and ε are, the more accurate the $\mathcal{R}(\mathbf{x})$. In particular, when δ and ε decrease to zero, then MRIP degenerates to the classical likelihood function denoting the probability of model prediction being the same as the actual observation given the input $\mathbf{x}$, namely $P(\widehat{y} = y|\mathbf{x}\,)$. In addition, an extra parameter c can be introduced to specify the adequacy requirement, such that we accept the ML model prediction only when $P(|y^* - \widehat{y}^*| \le \varepsilon \,|\mathbf{x}^* \in B(\mathbf{x})\,) \ge c$.

Note that the proposed reliability indicator – MRIP – is able to capture the uncertainty associated with both the input $\mathbf{x}$ and the observed target y because the MRIP indicator $\mathcal{R}(\mathbf{x})$ considers the variation associated with the observed difference $|\widehat{y} - y|$ when the input $\mathbf{x}$ varies subject to a predefined distance constraint δ. As multiple neighbors $\mathbf{x}^* \in B(\mathbf{x})$ encompassing $\mathbf{x}$ are considered in the developed MRIP estimator, the uncertainty pertaining to $\mathbf{x}$ is realized in these neighbor instances $\mathbf{x}^*$. Afterwards, the uncertainty associated with $\mathbf{x}$ gets propagated and translated into the model prediction. Likewise, if uncertainty arises in y, such uncertainty gets manifested in the trained ML model directly as the target y used to train ML model is associated with noise. Hence, the variation of model prediction $\widehat{y}^*$ on $\mathbf{x}^*$ could approximate such uncertainty as long as the sample $\mathbf{x}^*$ is within a narrow distance away from $\mathbf{x}$.[3] Other methods, such as the RUE method developed by [23], is only able to capture either the uncertainty in the feature space $\mathbf{x}$ or the uncertainty associated with the target output, but they are unable to consider the uncertainty related to both the input $\mathbf{x}$ and target outputs y

[3] If the noise associated with y is homoscedastic, then such uncertainty is reflected in the variation of model predictions if we vary $\mathbf{x}$. If the noise is heterogeneous, the uncertainty can be treated as homoscedastic as long as we vary $\mathbf{x}$ with a narrow range.

simultaneously. We will demonstrate the advantage of the proposed MRIP indicator in uncertainty quantification over RUE in Section 4.

3.1.2. MRIP estimation

In terms of implementation, after the ML model $\widehat{y} = \widehat{f}(\mathbf{x}, \boldsymbol{\theta})$ is trained, we use a sampling approach to estimate the reliability of ML model prediction on any $\mathbf{x}$. To derive the MRIP value for a given input $\mathbf{x}$, we randomly generate M samples from $\mathbf{x}$'s neighbors $\mathrm{B}(\mathbf{x}) = \{\mathbf{x}^* \mid \|\mathbf{x}^* - \mathbf{x}\| \le \delta\}$ subject to the distance constraint $\|\mathbf{x}^* - \mathbf{x}\| \le \delta$. Afterwards, we count the number of samples with distance between the ML model prediction $\widehat{y}^*$ and the actual observation y^* associated with the randomly generated sample $\mathbf{x}^*$ less than the predefined tolerance ε. Suppose only m out of the M samples have observed difference within the predefined tolerance ε, then the MRIP for the ML model prediction for the individual input $\mathbf{x}$ is calculated as follows:

$$\mathcal{R}(\mathbf{x}) = \frac{m}{M}. \tag{2}$$

The MRIP indicator $\mathcal{R}(\mathbf{x})$ is a local reliability indicator that provides a direct, objective, and quantitative measure of the "reliability" or the probability of success of the ML model with respect to its prediction for the individual input $\mathbf{x}$, where the local information associated with the input x and the ML model $\widehat{y} = \widehat{f}(\mathbf{x})$ is fully exploited. Take the data shown in Fig. 1(a) as an example. The values of target variable are generated according to the following function:

$$y = 2 + 4.26 \times \left(e^{-\mathbf{x}} - 4 \times e^{-2x} + 3 \times e^{-3\mathbf{x}}\right) + e(x), \tag{3}$$

where $e(x)$ is the noise function with a zero mean and $e(x) \sim \mathcal{N}(0, \sigma(x))$, where

$$\sigma(x) = -0.1 \times \left(e^{-18x} - e^{-4x}\right). \tag{4}$$

Clearly, the noise function $e(x)$ has a heteroscedastic variance as the variance $\sigma(x)$ of the function varies with the specific value of x. For the sake of demonstration, we generate 1000 points from the interval $[0, 2]$ uniformly, of which 800 points are used to train a support vector regression (SVR) model and 200 points are used to test the performance of the trained SVR model. Fig. 1(a) shows the 1000 randomly generated data points, and variation in the spread of y in Fig. 1(a) explicitly exhibits the heteroscedastic nature of the underlying noise function. By grid search cross validation, we find out that the SVR model achieves the best performance when epsilon is 0.001, the regularization parameter is 100 and rbf is adopted as the kernel function. Fig. (1)(b) illustrates the performance of the trained SVR model in fitting the 200 test data points. As can be observed, the SVR model captures the changing trend of the underlying function quite well, whereas it fails to account for the heterogeneous variance in the data. In particular, the error $|\widehat{y} - y|$ of the SVR model reaches the maximum value 0.488 when $x = 0.24$ as illustrated in Fig. 1(b). Without the MRIP indicator, there is no informative measure to indicate whether the SVR model prediction at $x = 0.24$ is reliable or not. To estimate MRIP, we set ε as 0.04, δ as 0.06 and N as 10,000. In practice, the values of ε and δ can be set according to application-specific requirements, and we will show how to choose them in Section 4.3.2. Assume that the underlying function shown in Eq. (3) is known to us, Fig. 1(c) illustrates the histogram of $y - \widehat{y}$ corresponding to the 10,000 randomly generated samples when $\mathbf{x}$ is 1.043. Among the 10,000 samples, only 6910 samples have $|y^* - \widehat{y}^*|$ (see the samples within the two black dashed lines in Fig. 1(c)) less than the predefined threshold ε. Thus, the MRIP for $\mathbf{x}$ at 1.043 is 0.691. In a similar manner, the MRIP for other $\mathbf{x}$ can be derived accordingly.

In practice, the underlying process governing the data generation as given in Eqs. (3) and (4) is typically unknown. Consequently, it is impossible to follow the steps in the sampling-based approach to derive $\mathcal{R}(\mathbf{x})$ because we are unable to get the corresponding target value y for each neighboring point $\mathbf{x}^* \in B(\mathbf{x})$ around the input $\mathbf{x}$. To tackle this problem, we propose to use the neighbors around $\mathbf{x}$ that are available in the observed data to approximate $B(\mathbf{x})$ (note that the data points are still subject to the distance constraint as defined in Eq. (1), where δ has a value of 0.06 and its value is very small in comparison to the mean value of y – 1.782). Estimating $\mathcal{R}(\mathbf{x})$ from neighboring points explicitly highlights the value of the developed MRIP indicator in tackling the reliability of individual prediction produced by a ML model. Following this idea, the MRIP for each input $\mathbf{x}$ can be approximated accordingly. Fig. 1 (d) illustrates the relationship between the sampling approach-based MRIP value $\mathcal{R}(\mathbf{x})$ and the approximated MRIP value $\widetilde{\mathcal{R}}(\mathbf{x})$. As can be observed, $\widetilde{\mathcal{R}}(\mathbf{x})$ fluctuates at a much higher magnitude than $\mathcal{R}(\mathbf{x})$. The high magnitude of fluctuation is understandable because now we rely on a limited number of neighboring samples encompassing the input $\mathbf{x}$ to estimate its MRIP value. Nevertheless, the overall changing trend of MRIP of $\widetilde{\mathcal{R}}(\mathbf{x})$ is still in good agreement with that of $\mathcal{R}(\mathbf{x})$ when $\mathbf{x}$ varies within the range $[0, 2]$. From a quantitative point of view, $\mathcal{R}(\mathbf{x})$ and $\widetilde{\mathcal{R}}(\mathbf{x})$ has a correlation of 0.984, which partially reflects the efficacy of approximating $\mathcal{R}(\mathbf{x})$ via neighboring points.

As an extension to this computational study, we consider another case that the ground truth relationship between x and y remains the same as defined in Eq. (3), but the noise function follows a log-normal distribution described as below:

$$e(x) \sim \mathrm{Lognormal}\left(0, -0.3 \times \left(e^{-18x} - e^{-4x}\right)\right). \tag{5}$$

Following the same procedure (e.g., data split, hyperparameter tuning), we train a random forest consisting of 1000 base estimators to quantify the uncertainty in random forest's estimations. Fig. 2 compares the performance of random forest with the proposed approach: Fig. 2(b) visualizes the relationship between $|y - \widehat{y}|$ and $\sigma(\widehat{y})$, and Fig. 2(c) shows the relationship between $|y - \widehat{y}|$ and the corresponding MRIP value $\mathcal{R}(\mathbf{x})$. To compare the usefulness of each indicator at signaling unreliable model predictions, we examine the correlation between the prediction error and corresponding reliability indicator. Quantitatively, $|y - \widehat{y}|$ and $\sigma(\widehat{y})$ has a correlation of 0.6438 while $|y - \widehat{y}|$ and $\mathcal{R}(\mathbf{x})$ has a correlation

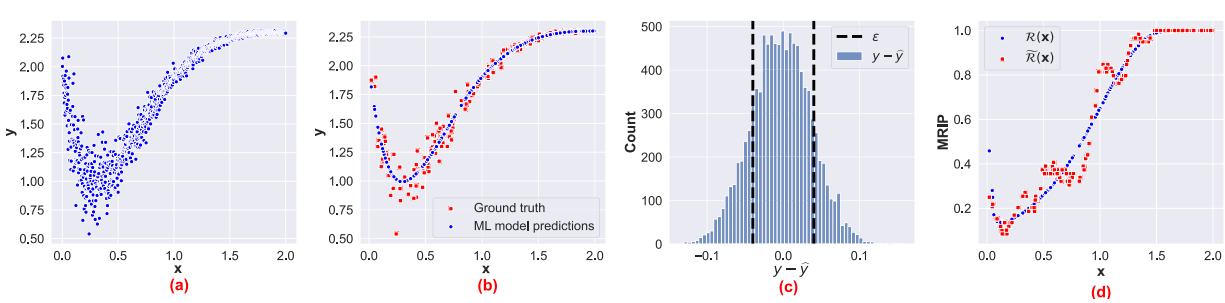


Fig. 1. Illustration of the MRIP indicator $\mathcal{R}(x)$ developed in this paper. (a) 1000 data points are randomly generated from the function $y = 2 + 4.26 \times \left(e^{-x} - 4 \times e^{-2x} + 3 \times e^{-3x}\right) + e(x)$, where $e(x) \sim \mathcal{N}(0, \sigma(\mathbf{x}))$, $\sigma(x) = -0.1 \times \left(e^{-18x} - e^{-4x}\right)$ and $x \in [0, 2]$. (b) The performance of a trained SVR model on 200 test data points. (c) Histogram of $y^* - \widehat{y}^*$ of 10,000 randomly generated samples when estimating the MRIP for the individual input x at 1.043. (d) Changing trend of $\mathcal{R}(x)$ versus $\widetilde{\mathcal{R}}(x)$.

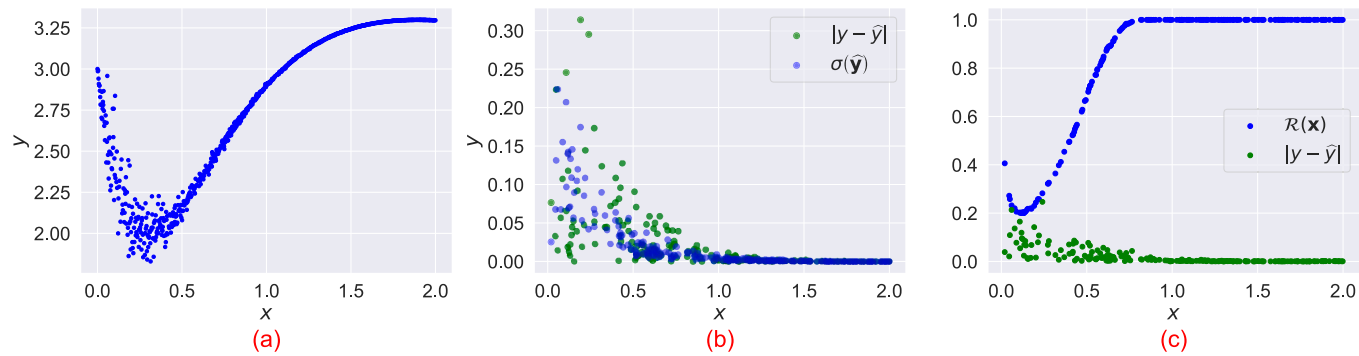


Fig. 2. Comparison of uncertainty estimation between the proposed method and random forest. (a) 1000 random samples are generated, (b) prediction intervals generated by random forest, (c) MRIP estimation by the proposed approach.

of 0.7377. The nice alignment between $|y-\hat{y}|$ and $\mathcal{R}(x)$ suggests that the estimated MRIP value is much more informative than $\sigma(\hat{y})$ in reflecting the prediction error $|y-\hat{y}|$. Such a side-by-side comparison highlights the advantage of the developed MRIP indicator over random forest in characterizing different types of noise in the underlying function.

Another appealing feature is that the proposed MRIP indicator $\mathcal{R}(\mathbf{x})$ is model agnostic. In other words, the MRIP indicator $\mathcal{R}(\mathbf{x})$ can be combined with any type of ML model ranging from random forest, neural network to logistic regression and support vector machine. In addition, the data-driven nature of the MRIP indicator $\mathcal{R}(\mathbf{x})$ frees us from making any statistical assumptions about the parameters and structures of the ML model as well as the underlying noise. Last but not least, the MRIP indicator considers the variation pertaining to both the model input and the ML model, thus it allows us to characterize the heterogeneous stochasticity in a straightforward manner.

3.2. Two-stage machine learning-based framework for MRIP estimation

Although MRIP offers a direct and objective measure for the reliability of ML model with respect to each individual prediction, the method of estimating $\mathcal{R}(\mathbf{x})$ from its corresponding neighbors suffers from two major drawbacks: (I) the method might fail to produce an objective reliability estimate if the neighbors are scarce around the input $\mathbf{x}$; (II) in the high dimensional space, if the input data $\mathbf{x}$ has a large number of features (e.g., 100 features), it might get computationally expensive to identify the neighbors around the input $\mathbf{x}$ subject to the distance constraint. To address these issues, we propose a two-stage ML-based framework to learn the relationship between input data $\mathbf{x}$ and the approximated MRIP $\widetilde{\mathcal{R}}(\mathbf{x})$. Once the ML model is developed, it can be leveraged to predict the output $\hat{y}$ corresponding to any neighboring point around the input $\mathbf{x}$. In this subsection, we introduce the two-stage ML-based approach for MRIP estimation.

As a quick overview, Fig. 3 illustrates the detailed procedures in the two-stage ML-based framework for MRIP estimation. Consider a dataset $(\mathbf{X},\mathbf{Y})$, the dataset is randomly split into two parts $(\mathbf{X}_{\text{train}},\mathbf{Y}_{\text{train}})$ and $(\mathbf{X}_{\text{test}},\mathbf{Y}_{\text{test}})$, where the former dataset $(\mathbf{X}_{\text{train}},\mathbf{Y}_{\text{train}})$ is used to train a ML model $\hat{y}=\hat{f}(\mathbf{x})$. In both datasets, $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$ denote the feature vectors corresponding to the training data and the testing data, respectively, while $\mathbf{Y}_{\text{train}}$ and $\mathbf{Y}_{\text{test}}$ are the target variable values the ML model aims to predict. After the ML model $\hat{y}=\hat{f}(\mathbf{x},\boldsymbol{\theta})$ is properly trained, then we derive the MRIP value for any $\mathbf{x}\in\mathbf{X}_{\text{test}}$ using the approximation method described in the previous subsection. As a result, we obtain a pair of data $(\mathbf{x},\widetilde{\mathcal{R}}(\mathbf{x}))$ consisting of the input data $\mathbf{x}$ and the corresponding MRIP indicator $\widetilde{\mathcal{R}}(\mathbf{x})$ for any $\mathbf{x}\in\mathbf{X}_{\text{test}}$. The goal of the first step is to derive the MRIP values for any $\mathbf{x}\in\mathbf{X}_{\text{test}}$ to prepare for the training of the surrogate model for MRIP estimation in the next stage.

Now, given the derived dataset $(\mathbf{x},\widetilde{\mathcal{R}}(\mathbf{x}))$ for any $\mathbf{x}\in\mathbf{X}_{\text{test}}$, we split it into two parts: $(\mathbf{X}^{\text{MRIP}}_{\text{train}},\widetilde{\mathcal{R}}_{\text{train}})$ and $(\mathbf{X}^{\text{MRIP}}_{\text{test}},\widetilde{\mathcal{R}}_{\text{test}})$ for training the MRIP surrogate model and validating its performance. Specifically, the dataset $(\mathbf{X}^{\text{MRIP}}_{\text{train}},\widetilde{\mathcal{R}}_{\text{train}})$ is used to train a surrogate model $\hat{\mathcal{R}}(\mathbf{x})=\hat{g}(\mathbf{x},\boldsymbol{\omega})$ to replace the original expensive neighboring points-based approximation approach for estimating the MRIP of the ML model $\hat{y}=\hat{f}(\mathbf{x},\boldsymbol{\theta})$ with respect to its prediction on the individual input $\mathbf{x}$, where $\boldsymbol{\omega}$ denotes the parameters of the surrogate model. After the surrogate model $\hat{\mathcal{R}}(\mathbf{x})=\hat{g}(\mathbf{x},\boldsymbol{\omega})$ is trained, it can be used to estimate the MRIP $\hat{\mathcal{R}}(\mathbf{x})$ for any $\mathbf{x}\in\mathbf{X}^{\text{MRIP}}_{\text{test}}$. The performance of the surrogate model $\hat{\mathcal{R}}(\mathbf{x})=\hat{g}(\mathbf{x},\boldsymbol{\omega})$ can be verified by comparing the prediction of the surrogate model $\hat{\mathcal{R}}(\mathbf{x})$ with the approximated MRIP value $\widetilde{\mathcal{R}}(\mathbf{x})$ for any $\mathbf{x}\in\mathbf{X}^{\text{MRIP}}_{\text{test}}$. Note that the

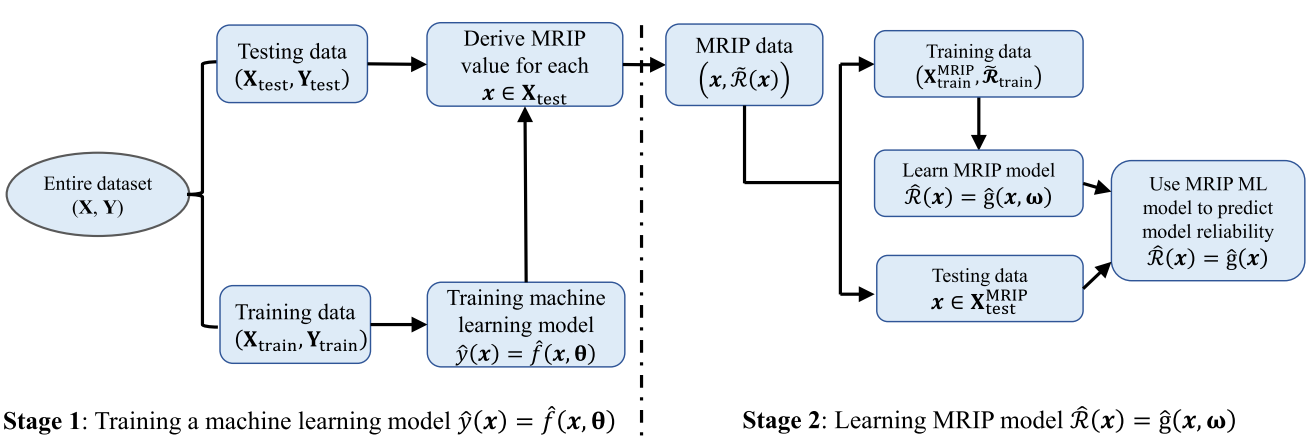


Fig. 3. Two-stage ML-based framework for estimating model reliability for individual prediction (MRIP).

surrogate model $\widehat{\mathcal{R}} = \widehat{g}(\mathbf{x}, \boldsymbol{\omega})$ can be trained with any ML model as demonstrated in the next section.

It is worth mentioning that the surrogate model $\widehat{\mathcal{R}}(\mathbf{x}) = \widehat{g}(\mathbf{x}, \boldsymbol{\omega})$ is different from the neighboring points-based approximation method in nature as it is able to estimate the MRIP value for any $\mathbf{x} \in \mathbf{X}$ instantly. The enormous reduction in the computational time when estimating MRIP has the potential of significantly facilitating the adoption of the reliability indicator MRIP in time-sensitive domains. In summary, the integration of the ML model $\widehat{y}(\mathbf{x}) = \widehat{f}(\mathbf{x}, \boldsymbol{\theta})$ and the surrogate model $\widehat{\mathcal{R}}(\mathbf{x}) = \widehat{g}(\mathbf{x}, \boldsymbol{\omega})$ offers an intuitive and scalable approach to provide an informative reliability estimate of the ML model with respect to its prediction for any individual input $\mathbf{x}$. The outcome of doing this is that, for each input $\mathbf{x}$, we obtain two quantities: the prediction of the ML model $\widehat{y}(\mathbf{x})$ and the MRIP indicator $\widehat{\mathcal{R}}(\mathbf{x})$. The estimated reliability indicator $\widehat{\mathcal{R}}(\mathbf{x})$ provides valuable insights about the ML model predictive performance specific to each input $\mathbf{x}$, and such knowledge is critical for supporting risk-informed decisions. With $\widehat{\mathcal{R}}(\mathbf{x})$, ML model predictions that are deemed unreliable can be identified beforehand, and this in turn can help prevent ML systems from causing unintended outcomes or harmful behaviour in practice.

3.3. Information gain-driven reliability threshold determination

Now each input $\mathbf{x} \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$ has a pair of quantities $(\widehat{y}(\mathbf{x}), \widehat{\mathcal{R}}(\mathbf{x}))$ to indicate the ML model performance. The next question is how to identify an appropriate value from the set $\widehat{\mathcal{R}}_{\text{test}}$ to act as the threshold to decide when to abstain from depending on the ML model prediction and request the intervention of human experts when the MRIP indicator $\widehat{\mathcal{R}}(\mathbf{x})$ is below a certain value. In this paper, we develop an information gain-driven approach to determine the threshold value of $\widehat{\mathcal{R}}(\mathbf{x})$. As illustrated in the stage 2 of Fig. 3, the MRIP indicator $\widehat{\mathcal{R}}(\mathbf{x})$ is derived for each input $\mathbf{x} \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$. Suppose λ denote the threshold parameter, we determine the value of λ by maximizing the information gain of the split [48], where Shannon's entropy is leveraged to measure the amount of information that is gained due to the split [49]. More specifically, suppose $\mathscr{K}$ denotes a discrete random variable with distribution given by:

$$Pr(\mathscr{K} = k_i) = p_i \quad \text{for} \quad i = 1, 2, \cdots, n. \tag{6}$$

The amount of information in the probability distribution $\mathscr{K}$ can be quantified using Shannon's entropy, which is widely used in information theory [49–51]. Mathematically, Shannon's entropy is defined as follows:

$$H(\mathscr{K}) = -\sum_{i=1}^{n} p(k_i) log\, p(k_i). \tag{7}$$

Afterwards, the information gain of the split is modeled as:

$$G(\mathscr{S}, \mathscr{T}) = H(\mathscr{S}) - H(\mathscr{T}), \tag{8}$$

where

$$\begin{aligned} H(\mathscr{S}) &= H(\widehat{\mathcal{R}}_{\text{test}}), \\ H(\mathscr{T}) &= \frac{\langle \widehat{\mathcal{R}}_{\text{test}}^{<} \rangle}{\langle \widehat{\mathcal{R}}_{\text{test}} \rangle} H(\widehat{\mathcal{R}}_{\text{test}}^{<}) + \frac{\langle \widehat{\mathcal{R}}_{\text{test}}^{>} \rangle}{\langle \widehat{\mathcal{R}}_{\text{test}} \rangle} H(\widehat{\mathcal{R}}_{\text{test}}^{>}). \end{aligned} \tag{9}$$

where $\widehat{\mathcal{R}}_{\text{test}}^{<} = \{\widehat{\mathcal{R}}(\mathbf{x}) : \widehat{\mathcal{R}}(\mathbf{x}) < \lambda\}$ for any $\mathbf{x} \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$, $\widehat{\mathcal{R}}_{\text{test}}^{>} = \{\widehat{\mathcal{R}}(\mathbf{x}) : \widehat{\mathcal{R}}(\mathbf{x}) \geq \lambda\}$ for any $\mathbf{x} \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$, the operator $\langle \widehat{\mathcal{R}}_{\text{test}} \rangle$ denotes the number of elements in the set $\widehat{\mathcal{R}}_{\text{test}}$.

For a fixed set $\mathscr{S}$, maximizing the objective function formulated in Eq. (8) is equivalent to minimizing the function $H(\mathscr{T})$. To maximize the utility of the MRIP indicator derived for the data $\mathbf{X}_{\text{test}}^{\text{MRIP}}$, the problem formulated in Eq. (8) recasts the determination of threshold value λ as an optimization problem, this provides a theoretically sound foundation to decide the value of λ. The threshold value obtained by solving the optimization problem can be used as an indicator to decide when to defer to a human expert for further examination and judgement. In practice, if users have a preferred reliability requirement, then the preferred reliability condition can override the value derived by solving the optimization problem.

4. Computational experiments

In this section, we first illustrate the advantage of the proposed MRIP indicator in signaling the reliability of individual ML prediction by comparing it against several prevailing reliability indicators in the literature. Next, we use eight datasets from the UCI machine learning repository to examine the performance of nine different ML models in MRIP estimation following the steps described in the two-stage ML-based approach.

4.1. Performance metrics

1. **RMSE-based metric**. If the actual MRIP value $\mathcal{R}(\mathbf{x})$ can be derived from the ML model using the sampling-based approach, then root mean squared error (RMSE)-based quantitative metric can be adopted to compare the MRIP estimator $\widetilde{\mathcal{R}}(\mathbf{x})$ derived from neighboring samples against the MRIP value $\mathcal{R}(\mathbf{x})$ that is extracted with the sampling-based approach. The RMSE-based quantitative metric will provide valuable information on the sufficiency of estimating $\mathcal{R}(\mathbf{x})$ from neighboring points. Mathematically, the RMSE-based metric is defined as $\sqrt{\frac{\sum_{i=1}^{K}(\mathcal{R}(\mathbf{x}) - \widetilde{\mathcal{R}}(\mathbf{x}))^2}{K}}$, $\forall \mathbf{x} \in \mathscr{X}$, where K indicates the number of elements in $\mathscr{X}$ and $\mathscr{X}$ refers to the set of dataset used to test the performance of $\widetilde{\mathcal{R}}(\mathbf{x})$ in approximating $\mathcal{R}(\mathbf{x})$.
2. **Correlation-based metric**: As illustrated in Fig. 3, the test data $\mathbf{X}_{\text{test}}^{\text{MRIP}}$ can be used to validate the effectiveness of the proposed MRIP indicator $\mathcal{R}(\mathbf{x})$ and the two stage ML-based framework in modeling the reliability of the ML models. Intuitively, a MRIP indicator should reasonably reflect the noise associated with the underlying function. From this standpoint, the MRIP indicator is expected to negatively correlate with the actual error $|y - \widehat{y}|$ of the ML model for the test dataset. Since our goal is to detect ML predictions violating the prediction accuracy requirement, having a high negative correlation between $|y - \widehat{y}|$ and the reliability estimator $\mathcal{R}(\mathbf{x})$ is a desirable property in the sense that low MRIP value can be used to identify individual ML prediction violating the predefined accuracy requirement $|y - \widehat{y}| > \varepsilon$.

   Correlation coefficient, as a widely adopted metric in the literature [19,20,44], is leveraged to quantitatively measure the relationship between $|y - \widehat{y}|$ and $\widehat{\mathcal{R}}(\mathbf{x})$ as estimated by the MRIP prediction model (see Fig. 3). As a measure of "alignment" to evaluate the performance of proposed MRIP indicator, the higher the correlation coefficient, the better the performance of the MRIP indicator. For each dataset, the reliability estimates over the testing data are correlated with their prediction error using the Pearson's correlation coefficient.

In the computational experiments, we use seven commonly used indicator-based reliability approaches as baselines for comparison: VarBag, VarA, VarP, MSE, as well as three sensitivity-based reliability indicators RE_1, RE_2, and RE_3 [52], see Table 1 for their detailed definitions. Unlike VarBag, VarA, VarP, and MSE, RE_1, RE_2, and RE_3 are three representative indicators in the class of transductive approaches, which is an approach that observes the output response with respect to small changes in the input data set. Considering the computational burden of transductive methods as reviewed earlier in Section 2, we only investigate the performance of the class of transductive-based indicators RE_1, RE_2, and RE_3 on selected datasets.

Table 1
Description of baseline methods considered in this paper.

Baseline	Description
VarBag	$\frac{1}{m}\sum_{j=1}^{m}(M_j-\widehat{y})^2, \widehat{y}=\frac{\sum_{j=1}^{m}M_j}{m}$; variance associated with M models' predictions on example $\mathbf{x}$, and the M models are trained with different data
VarA	$\frac{1}{n}\sum_{i=1}^{n}(\bar{y}-y_i)^2,\ \bar{y}=\frac{1}{n}\sum_{i=1}^{n}y_i$; VarA measures the variance of actual target variable values (y_i) associated with example $\mathbf{x}$'s n nearest neighbors
VarP	$\frac{1}{n}\sum_{i=1}^{n}(\bar{\widehat{y}}-\widehat{y}_i)^2,\ \bar{\widehat{y}}=\frac{1}{n}\sum_{i=1}^{n}\widehat{y}_i$; VarP quantifies the variance of the model's predictions ($\widehat{y}$) on n nearest neighbors around example $\mathbf{x}$
MSE	$\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i^2,\ \varepsilon_i=y_i-\widehat{y}_i$; MSE measures the mean squared error between the actual values (y_i) corresponding to example $\mathbf{x}$'s n nearest neighbors and the model predictions ($\widehat{y}_i$) on the n nearest neighbors around $\mathbf{x}$
RE$_1$	$\mathrm{RE}_1=\frac{\sum_{\varepsilon\in E}(K_\varepsilon-K_{-\varepsilon})}{\lvert E\rvert}$, where K_ε and $K_{-\varepsilon}$ denote the predictions of two sensitivity models on $\mathbf{x}$. Specifically, K_ε and $K_{-\varepsilon}$ denote the predictions of sensitivity models trained with a modified training set expanded with an additional learning example: in the case of $K_{-\varepsilon}$, we add an additional sample $(\mathbf{x}, y-\varepsilon)$ into the training data; in the case of K_ε, we add an additional sample $(\mathbf{x}, y+\varepsilon)$. Note ε denotes the level of random perturbation added to y.
RE$_2$	$\mathrm{RE}_2=\frac{\sum_{\varepsilon\in E}\lvert K_\varepsilon-K\rvert+\lvert K_{-\varepsilon}-K\rvert}{2\lvert E\rvert}$, where K_ε denotes the prediction of the sensitivity model on $\mathbf{x}$, and K denotes the prediction of the initial ML model on $\mathbf{x}$.
RE$_3$	$\mathrm{RE}_3=\frac{\sum_{\varepsilon\in E}(K_\varepsilon-K)+(K_{-\varepsilon}-K)}{2\lvert E\rvert}$, where K_ε and $K_{-\varepsilon}$ denote the predictions of two sensitivity models on $\mathbf{x}$.

4.2. MRIP illustration

MRIP demonstration: To illustrate the feature of MRIP in capturing both uncertainties associated with $\mathbf{x}$ and y, consider a function $y=\beta_1\left(1-exp^{(-\beta_2 x)}\right)$, where β_1 and β_2 take a value of 20 and 0.6, respectively. Suppose both the input x and model output y are associated with some random noise, then we represent the function as $y=\beta_1\left(1-exp^{(-\beta_2(x+\varepsilon_1))}\right)+\varepsilon_2$. Let $\varepsilon_1\sim\mathcal{N}(0.1,0.5), \varepsilon_2\sim\mathcal{N}(0,0.1)$, we randomly generate 40 points, use 80% of the data to train a ML model, and test the performance of the trained ML model with the remaining 20% data. We consider three different scenarios: (i) no noise at all in the data, (ii) noise is only associated with x; (iii) noise is present in both x and y. Intuitively, the uncertainty in model prediction is expected to increase after more and more random noise are injected into the function. Fig. 4 compares the uncertainty estimated by the RUE method and the MRIP value at each test point. As shown in Fig. 4(a), there is no significant difference regarding the uncertainty estimated by RUE method in the three respective scenarios. However, it can be observed from Fig. 4(b) that the MRIP value $\mathcal{R}(\mathbf{x})$ drops down after noise in x is injected into the function, and its value further decreases when noise appears in both x and y. Such an example clearly highlights the advantageous feature of the developed MRIP indicator in capturing the uncertainty associated with both the input x and the observed target y.

MRIP verification: To verify the effectiveness of the developed MRIP indicator using the two performance metrics defined earlier, we set up a computational experiment to assess the performance of different measures in quantifying the reliability of individual ML prediction. Suppose $\mathbf{x}$ and y follow a jointly normal distribution with mean and covariance matrix defined as below:

$$\boldsymbol{\mu}=\begin{bmatrix}\mu_y\\ \boldsymbol{\mu}_x\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix},\quad \boldsymbol{\Sigma}=\begin{bmatrix}\sigma_y^2 & 0.3 & 0.2\\ 0.3 & 1 & 0.2\\ 0.2 & 0.2 & 1\end{bmatrix}, \tag{10}$$

where σ_y denotes the standard deviation of target variable y, and σ_y^2 takes a value of 0.34.

By generating random samples with the multivariate Gaussian distribution defined above, we formulate the regression problem as predicting the value of y from the 2-dimensional input $\mathbf{x}$. Representing the relationship between $\mathbf{x}$ and y with the multivariate normal distribution enables to estimate $P(|y-\widehat{y}|\le\varepsilon)$ analytically. Next, we compare the performance of MRIP estimation with the developed method against other state-of-the-art reliability measures, including VarBag, VarA, VarP, VarMSE, and three transductive-based methods denoted by RE$_1$, RE$_2$, and RE$_3$.[4] To this end, 100 random samples are generated from the multivariate Gaussian distribution, and 80% of the data is used to tune the parameters of the SVR model and the remaining 20% data is used to validate the performance of reliability estimation. Note that SVR is used as the ML model across all the methods. Fig. 5(a) shows the reliability estimated by each method, and the developed MRIP indicator demonstrates a superior performance in the reliability estimation pertaining to each individual SVR prediction compared to other alternatives. Quantitatively, the estimated MRIP and the ground truth value has an average MSE of 0.02, and outperforms all the other reliability indicators in the literature. Table 2 reports the computational time of deriving the value of each reliability indicator. Among all the methods, transductive-based methods are the most time-consuming with computational time larger than the others by at least two orders of magnitude.

Next, we investigate the robustness of the developed MRIP indicator by varying the dataset size and the variation associated with the target variable y. Specifically, we gradually increase the value of σ_y^2 from 0.34 to 0.84 with a step size of 0.1 while keeping the other entries in the covariance matrix $\sum$ the same. As shown in Fig. 5(b), the proposed MRIP indicator is always closest to the ground truth values when compared with the other state-of-the-art reliability measures. Note that ε and σ are fixed at 0.1 and 0.05, respectively. Next, we investigate the impact of the number of samples on the reliability estimation of $\mathcal{R}(x)$, where the number of training samples is increased from 48 (60*0.8) to 480 (600*0.8) gradually. As shown in Fig. 5(c), the ground truth and the value of $\mathcal{R}(x)$ get boosted in a steady manner while $\mathcal{R}(x)$ and the ground truth values remain consistent. The computational comparison and robustness study suggests that $\mathcal{R}(x)$ is a good pseudo measure in estimating the reliability associated with each individual ML prediction.

In what follows, we consider a slightly more complex numerical example to verify the usefulness of the proposed MRIP indicator. In this example, y and $\mathbf{x}$ follows a nonlinear relationship $\mathbf{y}=4\mathbf{x}_{:,1}^2-3\mathbf{x}_{:,2}^2-6\mathbf{x}_{:,3}+4\mathbf{x}_{:,4}+8\mathbf{x}_{:,5}$, where $\mathbf{x}_{:,i}$ is the i-th column of $\mathbf{x}$ and $\mathbf{x}_{:,1},\mathbf{x}_{:,2},\mathbf{x}_{:,3},\mathbf{x}_{:,4},\mathbf{x}_{:,5}\sim\mathcal{N}(0,1)$. Clearly, the expected value and variance of y for any given $\mathbf{x}$ can be derived in a closed form. Next, we follow the same experimental configuration to derive the value of each reliability estimator and compare their values against the ground truth. As shown in Fig. 6(a), MRIP demonstrates superior performance than other reliability estimators. As displayed in Fig. 6(b), most MRIP values clusters around the straight line with an angle of 45 degree. In other words, the MRIP values highly align with the ground truth values, and it thus serves as a reliable and informative indicator to accurately reflect the reliability associated with each individual prediction. Similarly, we also compare the computational time of all the methods. As shown in Table 3, as expected, the transductive-based methods (e.g., RE$_1$, RE$_2$, and RE$_3$) are the most computational demanding among all the methods.

The above computational study on the two synthetic datasets showcases the high computational overhead of transductive-based reliability indicators due to the need to train individual sensitivity regression models every time the training set is expanded with an additional learning instance. In other words, if there are l data points in $\mathbf{X}_{\text{test}}^{\text{MRIP}}$, we need to train l sensitivity regression models with each model corresponding to one test sample in $\mathbf{X}_{\text{test}}^{\text{MRIP}}$, let alone the hyperparameter

[4] Refer to the appendix for the detailed implementation of each baseline method.

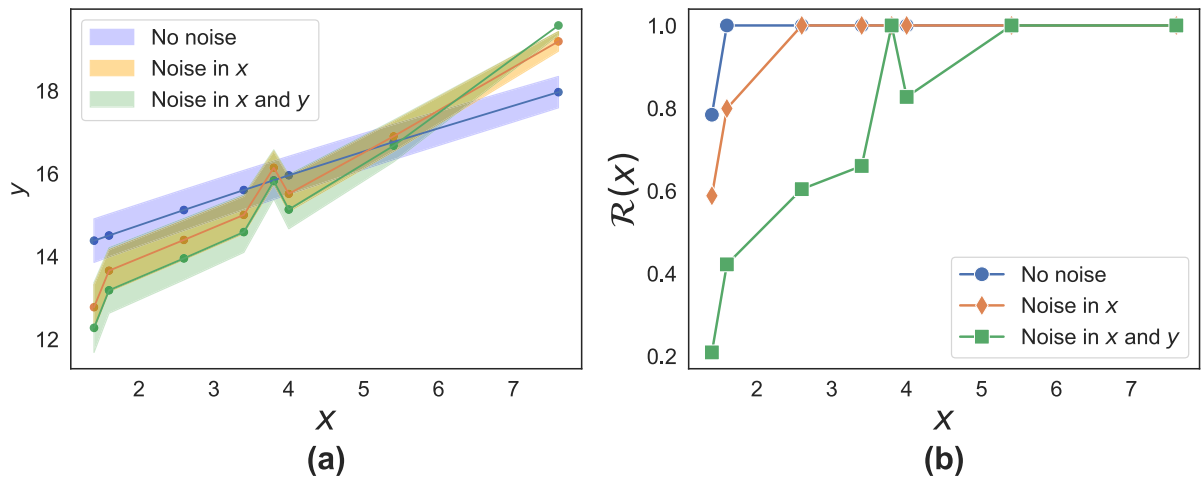
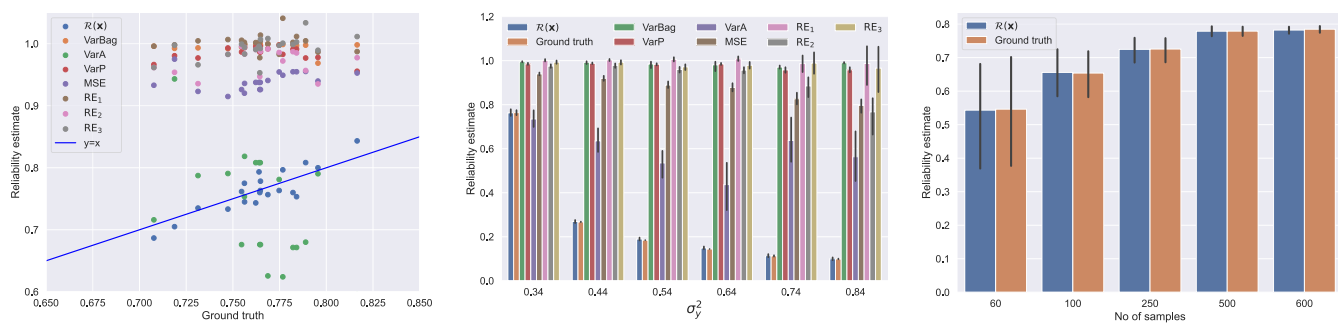


Fig. 4. (a) RUE-based uncertainty estimation for the three considered scenarios. (b) MRIP estimated by the proposed method. Note that δ and ε take a value of 0.15 and 0.005 when deriving the MRIP value.



(a) Comparison of reliability estimation by different methods

(b) $\mathcal{R}(x)$ vs σ_y^2

(c) $\mathcal{R}(x)$ vs the number of samples

Fig. 5. Performance comparison of reliability estimation and the relationship between $\mathcal{R}(x)$ and σ_y^2 as well as the number of training samples. Note that δ and ε take a value of 0.1 and 0.05 when deriving the MRIP value.

Table 2
Computational time comparison of different methods over the test data in the first numerical example.

Method	MRIP	VarBag	VarA	VarP	VarMSE	RE_1	RE_2	RE_3
Computational time (Unit: Seconds)	0.466	4.980	0.008	0.023	0.025	300	298	302

tuning needed for training each sensitivity model. The large number of sensitivity regression models to be trained, the high computational burden incurred by hyperparameter turning together with the number of cross validations makes it unrealistic to benchmark the performance of the developed approach against the transductive-based methods across all the datasets. As a result, we only investigate the performance of the three transductive-based reliability indicators on selected datasets.

Unlike the transductive-based reliability indicators, the proposed method for MRIP estimation only needs to train two models in total: one model is employed for generating $\widehat{y}(\mathbf{x})$ for the input $\mathbf{x}$, and the other model is used for estimating MRIP value $\widetilde{\mathcal{R}}(\mathbf{x})$ associated with the input $\mathbf{x}$. In other words, the second ML model, after training, can generate the reliability estimations for all the $x \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$ in one shot. That is, in the second stage, we do not need to train an individual ML model for each $x \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$. Instead, we train a single ML model to generate MRIP values for all the $x \in \mathbf{X}_{\text{test}}^{\text{MRIP}}$. Such a difference makes the proposed method more computationally efficient compared to the transductive-based methods.

4.3. Computational study

As mentioned earlier, the computational study consists of two phases. In the first phase, we consider two synthetic functions, where the ground truth relationship between $\mathbf{x}$ and y as well as the underlying function to generate random noise is known to us. In this setting, we leverage the method described in Section 3.1.2 to derive the MRIP value $\mathcal{R}(\mathbf{x})$ and $\widetilde{\mathcal{R}}(\mathbf{x})$ for assessing the quality and sufficiency of the developed reliability indicator in reflecting the likelihood of model prediction error under noisy environment using correlation and RMSE-based metrics. In contrast, in the second phase of computational study, since only the ground truth value y is available and there is no way to estimate $\mathcal{R}(\mathbf{x})$. Therefore, we rely on the correlation between $|y - \widehat{y}|$ and $\widehat{\mathcal{R}}(\mathbf{x})$ to evaluate the quality of the reliability indicator.

4.3.1. Phase 1

In the first phase of computational study, we concentrate on examining the relationship between $\mathcal{R}(\mathbf{x})$ and $\widetilde{\mathcal{R}}(\mathbf{x})$ to evaluate the perfor-

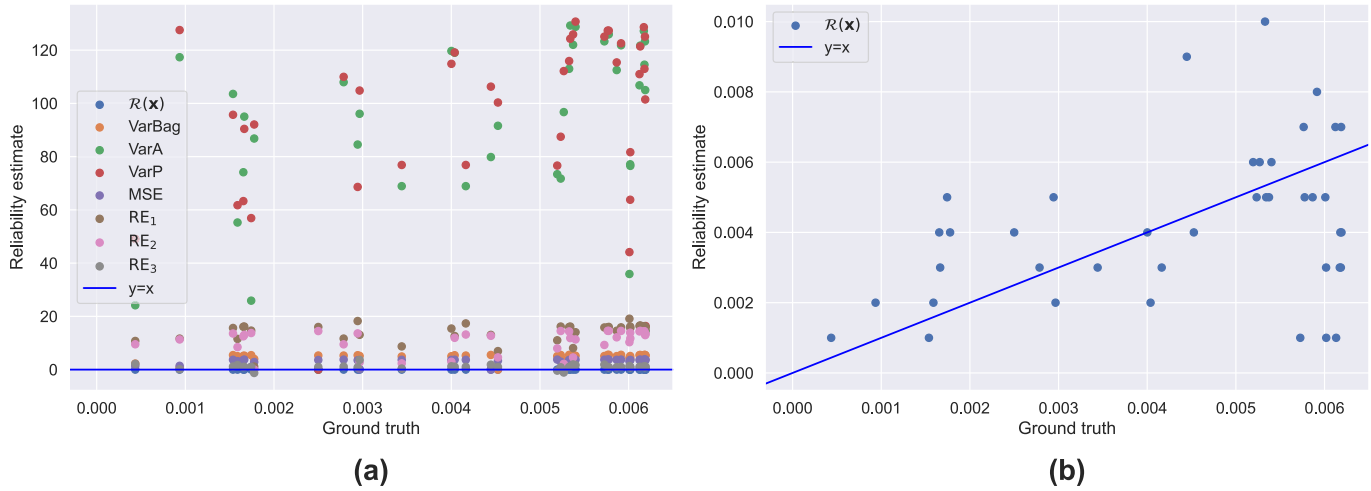


Fig. 6. Performance comparison of reliability indicators. Note that δ and ε take a value of 0.1 and 0.05 when deriving the MRIP value.

Table 3
Computational time comparison of different methods over the test data in the nonlinear numerical example.

Method	MRIP	VarBag	VarA	VarP	VarMSE	RE_1	RE_2	RE_3
Computational time (Unit: Seconds)	1.380	3.500	0.012	0.043	0.040	490	480	489

mance of the developed approach, where $\mathcal{R}(\mathbf{x})$ denotes the MRIP indicator derived using the sampling-based approach and $\widetilde{\mathcal{R}}(\mathbf{x})$ indicates the MRIP value approximated using the limited neighboring points around the input $\mathbf{x}$. To perform the experimental study, we assume that the underlying function to generate the random noise is known to us.

Table 4 displays the detailed information of the two synthetic functions. The two synthetic functions characterizes nonlinear relationships between $\mathbf{x}$ and y, where the ground truth functions are associated with different source of noise with varying noise magnitude. The two synthetic functions are designed to help to examine the sufficiency of the proposed MRIP indicator in reflecting the likelihood of model prediction error under noisy environment. As can be observed, the first synthetic function has an underlying noise pertaining to the observed target $y(x)$ and the noise follows a Gaussian distribution with zero mean and $\frac{x^2+sin(x)+2}{\tau}$ as variance. Note that the variance of the Gaussian distribution varies with the specific value of x, thus reflecting the heterogeneous nature of the underlying noise. To examine the performance of the developed MRIP indicator, we experiment with three different values of τ: 1, 5, 10. Obviously, the smaller the τ, the stronger the magnitude of the heterogeneous noise.

Unlike the first synthetic function, the second synthetic function has five dimensions with heterogeneous noises arising from both the input $\mathbf{x}$ and the target value $y(\mathbf{x})$. Specifically, we intentionally inject a heterogeneous noise pertaining to the input variable x_2 and $y(\mathbf{x})$, which are characterized by two random variables ϵ and γ, respectively. Both ϵ and γ follow Gaussian distributions with zero mean: ϵ has a variance of $\frac{x_2}{5}$ and γ has a variance of $\frac{0.0647\left(12+3x_1-3.5x_2^2+7.2x_3^3\right)(1+cos(4\pi x_4))*(1+0.8sin(3\pi x_5))}{5}$. Clearly, the second synthetic function is more challenging to model than the first synthetic function because of the inherent heterogeneous noise associated with both $\mathbf{x}$ and $y(\mathbf{x})$.

Note that the two experimental studies follow the same procedure. First of all, a fixed number of samples are generated uniformly subject to the range of each input variable. Next, the sampled data is randomly split into two parts: 80% of the data is used to train a SVR and the remaining 20% of the data is used to test the performance of estimating the MRIP using sampling-based approach and the neighboring points-based method. In both cases, we train a ε-SVR, where grid search cross validation is used to tune the parameters of SVR (e.g., type of kernel function, C, ϵ). Fig. 7(a) illustrates the correlation between $\widetilde{\mathcal{R}}(x)$ and $|y-\hat{y}|$. When τ takes the value of 1, 5, 10, $\widetilde{\mathcal{R}}(\mathbf{x})$ and $|y-\hat{y}|$ has a correlation of 0.423, 0.434, and 0.444, respectively. Fig. 7(b) illustrates the RMSE between $\mathcal{R}(x)$ and $\widetilde{\mathcal{R}}(x)$. The low RMSE suggests that estimating MRIP from neighboring points closely approximates the reliability of the SVR model with respect to its prediction for each individual input.

To deepen the understanding of the correlation between $\widetilde{\mathcal{R}}(x)$ and $|y-\hat{y}|$, Fig. 8 shows the relationship between $\widetilde{\mathcal{R}}(x)$ and $|y-\hat{y}|$ when τ takes different values. Different from the significant dispersion associated with $|y-\hat{y}|$, the MRIP indicator $\widetilde{\mathcal{R}}(x)$ generates a relatively stable upper bound to better reflect the reliability associated with each individual prediction due to its consideration of the random noise when estimating the individual model prediction reliability. While $|y-\hat{y}|$ might be low in some cases, the reliability estimation of individual ML prediction solely based on $|y-\hat{y}|$ might lead to an underestimation of

Table 4
Synthetic functions used to check the performance of the proposed MRIP metric in characterizing the heterogeneous noise.

	Expression of synthetic function
Synthetic function 1	$y(x) = x^2 + sin(x) + 2 + \gamma,\ x \in [-10, 10],\ \gamma \sim \mathcal{N}\left(0, \frac{x^2+sin(x)+2}{\tau}\right)$
Synthetic function 2	$y(x_1, x_2, x_3, x_4, x_5) = 0.0647\left(12 + 3x_1 - 3.5(x_2+\epsilon)^2 + 7.2x_3^3\right)(1 + cos(4\pi x_4))*(1 + 0.8sin(3\pi x_5)) + \gamma$, where $x_i \in [0,1], i = 1, \cdots, 5, \gamma \sim \mathcal{N}\left(0, \frac{0.0647(12+3x_1-3.5x_2^2+7.2x_3^3)(1+cos(4\pi x_4))*(1+0.8sin(3\pi x_5))}{5}\right)$ and $\epsilon \sim \mathcal{N}\left(0, \frac{x_2}{5}\right)$

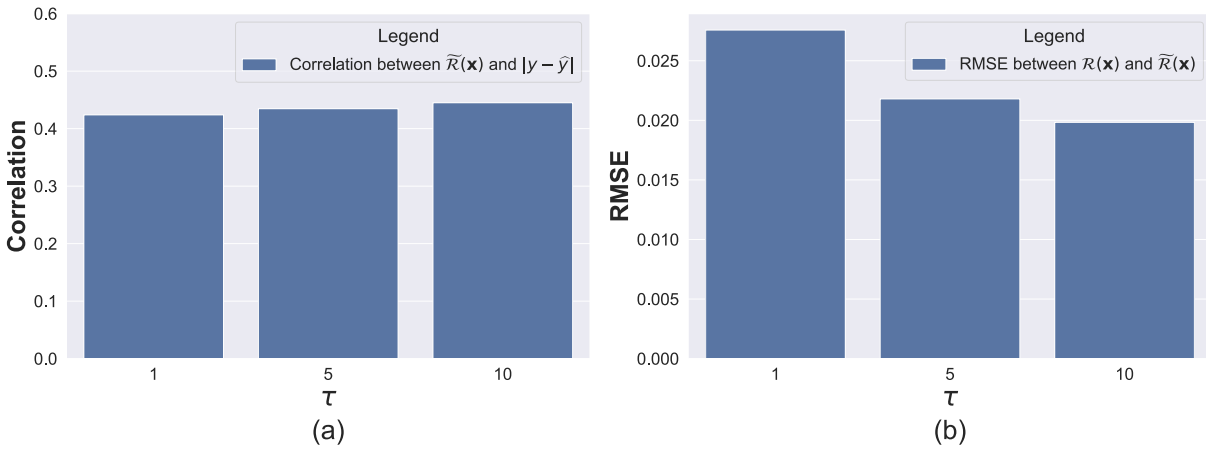


Fig. 7. Performance validation of the MRIP indicator in characterizing the heterogeneous noise associated with the first synthetic function. 2000 samples are uniformly generated to train and test the performance of the MRIP indicator. With respect to the sampling-based approach, 10,000 samples are randomly generated to estimate $\mathcal{R}(\mathbf{x})$ for each point in the test data. When deriving the value of the MRIP indicator, we set the values of ε and δ as 0.5 and 1, respectively. Note that $y(x)$ varies within the range $[-1.4, 120]$ when x is subject to the range $[-10, 10]$.

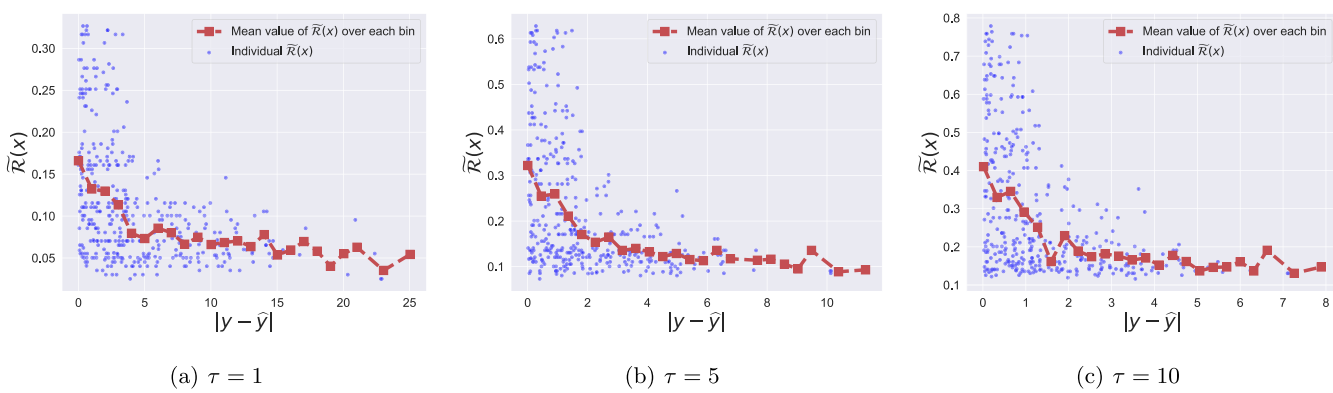


Fig. 8. Relationship between $\widetilde{\mathcal{R}}(x)$ and $|y-\hat{y}|$ when τ takes different values, where dashed lines indicate the changing trend of mean value $\widetilde{\mathcal{R}}(x)$ over each bin.

the actual risk pertaining to the individual ML prediction because the single value $|y-\hat{y}|$ is unable to fully accommodate the noise of random nature associated with the observation y. In other words, a low value of $|y-\hat{y}|$ does not imply that the individual ML prediction is free from risk. In particular, since the noise associated with y is irreducible in nature, it is reasonable to reflect these irreducible sources of noise in the reliability indicator to inform end users about the risk involved in each individual ML prediction that varies from case to case. As the developed approach provides a stable upper bound, we thus concentrate on analyzing the

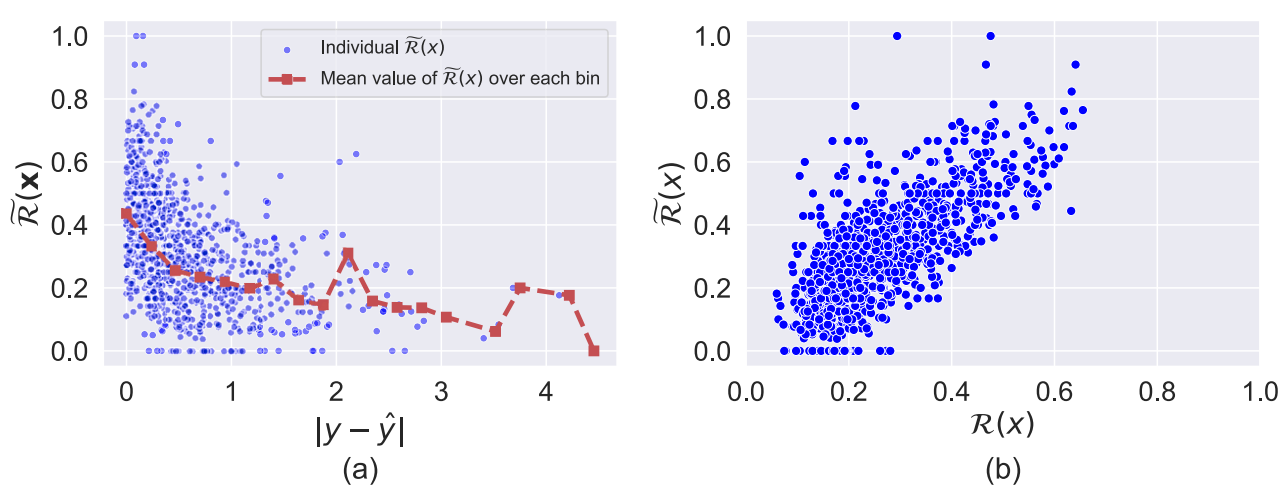


Fig. 9. Performance evaluation of MRIP indicator $\mathcal{R}(\mathbf{x})$ in characterizing the heterogeneous noise of the second synthetic function. 5000 samples of $\mathbf{x}$ are randomly generated. Note that $y(x)$ varies within the range $[0, 4.76]$ when $\mathbf{x}$ is subject to the range constraint $x_i \in [0,1]$, $i = 1, 2, \cdots, 5$. When deriving the value of $\mathcal{R}(\mathbf{x})$ for each $\mathbf{x}$ in the test data, 5000 samples are generated. In this case, ε and δ take a value of 0.2 and 0.25, respectively.

correlation between the upper bound and the absolute value of prediction error. To do this, we bin the data in accordance with the value of $|y-\widehat{y}|$ and compute the mean value of $\widetilde{\mathcal{R}}(x)$ associated with each bin, which are represented as dashed lines in Fig. 8. When τ takes the values 1, 5, and 10, the mean value of $\widetilde{\mathcal{R}}(x)$ over each bin has a correlation of 0.748, 0.719, and 0.808 with $|y-\widehat{y}|$, respectively, which in turn suggests that $\widetilde{\mathcal{R}}(x)$ accurately models the reliability of individual ML prediction.

With respect to the second synthetic function, Fig. 9 shows the performance of the developed MRIP indicator in capturing the composite heterogeneous noise arising from both the input variable $\mathbf{x}$ and the target $y(\mathbf{x})$. As shown in Fig. 9(a), $\widetilde{\mathcal{R}}(x)$ clearly captures the variation of $|y-\widehat{y}|$ in accordance with the increase in the standard deviation $\sigma(\mathbf{x})$ of the function. Overall, the mean value of $\widetilde{\mathcal{R}}(x)$ over each bin and $|y-\widehat{y}|$ has a high correlation of 0.764, which demonstrates the efficacy of the developed reliability indicator. Fig. 9(b) illustrates the relationship between $\mathcal{R}(x)$ and $\widetilde{\mathcal{R}}(x)$. Again, $\mathcal{R}(x)$ and $\widetilde{\mathcal{R}}(x)$ highly align with each other. Quantitatively, $\mathcal{R}(x)$ and $\widetilde{\mathcal{R}}(x)$ have an average RMSE of 0.127. The lower RMSE suggests that deriving the reliability of individual ML prediction using neighboring points provides a reasonable estimate of $\mathcal{R}(x)$.

Next, since the developed methodology is a two-stage learning in nature, we employ the MSE associated with the ML model at each stage to examine the reliability of the respective ML model. To this end, we further split the data points with $\mathcal{R}(\mathbf{x})$ derived into 80% training and 20% testing for the two synthetic functions, where another ML model is trained to predict the reliability estimate for each input $\mathbf{x}$. To compare the two ML models on a fair ground, we employ the same type of ML model and tune the same set of hyperparameters. Note that in the first stage, we report the MSE averaged over the samples randomly generated that are used to calculate the value of $\mathcal{R}(\mathbf{x})$. Fig. 10 compares the MSE for the test data associated with the first and second ML model. With respect to the first synthetic function, regardless of the value of τ, the first ML model has a much larger MSE than the second ML model. While in the case of second synthetic function, only 22.5% of data points have the MSE in the second ML model larger than that of the first ML model.

The above computational studies reveal that the proposed MRIP indicator shows a robust performance in recognizing samples that are intentionally generated with different magnitudes of noise. As a consequence, it has a great promise to serve as the safety guard to inform end users about the cases with high prediction error due to the underlying heterogeneous noise.

4.3.2. Phase 2

In the second phase of the computational study, we use several datasets from the UCI machine learning repository that are commonly used in the literature to validate the performance of the two stage ML-based approach in estimating the MRIP across several applications [53]. Table 5 describes the basic information of each dataset used in this paper, including the number of attributes, number of data points, the meaning of output and the range of the output.

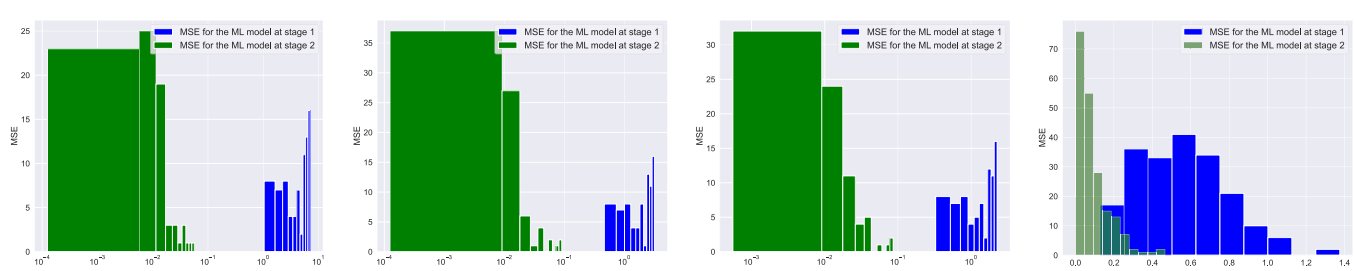
Note that all the computational experiments follow the same two-stage procedure. To be specific, in the first stage, we split the entire data into two parts randomly: 40% of data is used for training the ML model $\widehat{y}=\widehat{f}(\mathbf{x},\boldsymbol{\theta})$ and the remaining 60% of data is reserved for the MRIP model training and performance validation. In the second stage, we further split the 60% data into two equal parts randomly: 50% for learning the MRIP model $\widehat{\mathcal{R}}=\widehat{g}(\mathbf{x},\boldsymbol{\omega})$ and the remaining 50% of data to validate the performance of the learned MRIP model. Note that the same procedures for each ML model is repeated 20 times (each time, the random split varies, but within the same run, the split remains the same for all the ML models), and we report the performance of each ML model averaged over the 20 runs.

In the first stage, we train nine ML algorithms that are commonly used to build regression models, namely: decision tree (DT), linear regression (LR), ridge regression (RR), LASSO regression (LASSO), neural network (NN), random forest (RF), support vector regression (SVR), XGBoost regression (XGB) and k nearest neighbors (KNN). It can be observed that the developed MRIP indicator $\mathcal{R}(\mathbf{x})$ is general purpose because it can be combined with any ML model easily. In the second stage, we utilize the same nine ML models to investigate their performance in predicting the MRIP value on unobserved inputs. To avoid handpicking the value of ε and δ, we follow a simple logic to set the values of ε and δ in a unified way: the higher the variance of $\mathbf{Y}$, the higher the tolerance level ε; likewise, the more the number of features in $\mathbf{x}$, the larger the radius δ for neighboring points encompassing $\mathbf{x}$. Thus, we have the values of ε and δ defined as below:

$$\varepsilon=\frac{\sigma(\mathbf{Y})}{2},\qquad \delta=\sqrt{\Delta^{2^*}\lfloor\mathbf{x}\rfloor}, \tag{11}$$

where $\sigma(\mathbf{Y})$ denotes the standard deviation of the entire dataset of the target value $\mathbf{Y}$, Δ denotes the magnitude of change from the current value of $\mathbf{x}$ pertaining to each feature while we search for neighboring points (we set the value of Δ as 5% in this paper) to estimate MRIP, and $\lfloor\mathbf{x}\rfloor$ indicates the number of features in the input $\mathbf{x}$. Note that to make sure the values of all the features are in the same scale, we normalize the values of all the features within the range $[0,1]$ using the MinMaxScaler before searching for neighboring points to estimate the MRIP.

When training the ML models at both stages (e.g., $\widehat{y}=\widehat{f}(\mathbf{x},\boldsymbol{\theta})$, $\widehat{\mathcal{R}}(\mathbf{x})=\widehat{g}(\mathbf{x},\boldsymbol{\omega})$), grid search cross validation is used to tune the relevant parameters in each ML model. For example, the kernel function (e.g., rbf, linear, poly), the regularization parameter C and ϵ in the ϵ-SVR model; number of neighbors and weight function in KNN; the maximum depth and the minimum samples for splitting a node in DT and RF (number of estimators is another additional parameter to tune in RF); number of hidden layers, size of each hidden layer, activation function, batch size, learning rate in NN; and maximum depth, subsample ratio, learning rate, and minimum sum of instance weight in XGBoost. For the sake of completeness, we report the performance of each ML model across the eight datasets that is averaged over 20 runs in Table 6. It can



(a) $\tau=1$ (b) $\tau=5$ (c) $\tau=10$ (d) Synthetic function 2

Fig. 10. Performance comparison of ML model at each stage.

Table 5
Descriptions of datasets used in the computational experiments.

Dataset	No of attributes	No of observations	Output description	Output range
Boston housing	13	506	Median value of owner-occupied homes	[5.0, 50.0]
Power plant	4	9568	Electrical energy output of the plant	[420, 495]
Istanbul stock exchange	7	536	USD-based Istanbul stock exchange 100 index	[− 0.085, 0.101]
Parkinson's disease score	26	5875	Clinician's total UPDRS score	[7.0, 54.992]
Airfoil noise	6	1503	Sound pressure level	[103.38, 140.987]
Concrete compression strength	9	1030	Concrete compressive strength	[2.332, 82.599]
Naval propulsion	16	11,934	GT compressor decay state coefficient	[0.95, 1]
Auto MPG	8	398	Fuel efficiency	[9.0, 46.6]

Table 6
Performance comparison (e.g., $\sqrt{\frac{\sum_{i=1}^{n}(y_i-\widehat{y}_i)}{n}}$) of different ML models in predicting the target value of each UCI dataset.

Model	Boston Housing	Power plant	Istanbul stock exchange	Parkinson's disease score	Airfoil noise	Concrete compression strength	Naval propulsion	Auto MPG
DT	4.925	4.333	0.0167	3.757	3.381	8.138	0.0022	3.841
LR	4.965	4.595	**0.0141**	9.757	4.817	10.43	0.0058	3.477
RR	4.988	4.595	0.0142	9.741	4.818	10.43	0.0058	3.463
LASSO	5.037	4.594	0.0142	9.747	4.818	10.43	0.0075	3.436
NN	5.089	4.205	0.0144	6.070	2.817	6.673	0.0006	3.040
RF	3.874	3.693	0.0152	**2.540**	2.445	5.952	0.0015	3.060
SVR	5.240	4.092	0.0142	7.655	3.306	7.215	0.0003	**2.948**
XGB	**3.809**	**3.628**	0.0154	2.590	**2.192**	**5.253**	**0.0012**	3.040
KNN	6.721	3.950	0.0157	6.598	3.749	9.591	0.0018	3.130

be noticed that XGBoost achieves the lowest RMSE on several datasets, such as Boston housing, power plant, airfoil noise, concrete compression strength, and naval propulsion. While in the case of Istanbul stock exchange, LR achieves the best performance with a RMSE of 0.0141. When predicting Parkinson's score, RF has the lowest RMSE value followed by XGB. SVR outperforms other ML models when forecasting the auto MPG.

In this phase of computational study, since the ground truth function to generate the observed data is unknown, we are unable to leverage the sampling-based approach to extract the value of $\mathcal{R}(\mathbf{x})$. As a consequence, we can only rely on the correlation-based metric to validate the performance of the proposed methodology. Table 7 reports the performance of the combined ML models, where nine ML models are trained to predict the observed target y as well as the MRIP value associated with each predicted target $\widehat{y}$, respectively. Next, we use the correlation-based metric to examine the performance of the 81 models in total. In Table 7, the highest correlations generated by the combination of the best performed outcome model and the MRIP model are indicated in bold. As a quick overview, while the majority of the eight datasets have a correlation with values close to 0.5, the correlation for the dataset power plant and Istanbul stock exchange is relatively lower among all the datasets.

To further examine the performance of the developed approach for MRIP estimation, we compare it with VarBag, VarA, VarP and MSE across all the seven datasets. As transductive-based reliability indicators need to retrain the ML model from scratch every time the training set is expanded with an additional learning example, we thus only use SVR as the base model to verify the performance of RE_1, RE_21, and RE_3 in five small- to medium-size datasets. Note that in the case of VarBag, we train 10 ML models to estimate the prediction variance; we set the number of neighbors as 5 when deriving the reliability estimate for the other three baselines. In each run, we keep the data split for all the considered methods the same as that of MRIP. Table 7 displays the correlation between the prediction variance and the prediction error.[5] At a quick glance, it can be seen that the developed approach outperforms all the baselines in the correlation across all the considered datasets. Next, to evaluate the significance of correlation coefficient, we carry out the t-test for correlation coefficients using the p-value with a significance level $\alpha = 0.05$, where the null hypothesis is "there is not a significant linear relationship between MRIP and prediction error in the population". The hypothesis test results suggest that the best-performing models has a p value much lower than the significance level 0.05 for five datasets except naval propulsion, Istanbul stock exchange, and auto MPG. Since the correlation coefficient is significantly different from zero on these five datasets, we thus reject the null hypothesis and conclude that there is a significant linear relationship between $\widetilde{\mathcal{R}}(\mathbf{x})$ and $|y-\widehat{y}|$.

Note that the performance of MRIP estimation is an aggregated effect of multiple factors, such as the distribution of the data, the learning capacity of ML model, and the data sparseness. Both the ML model for outcome prediction and the second ML model for MRIP prediction have an essential impact on the correlation between $\widetilde{\mathcal{R}}(\mathbf{x})$ and $|y-\widehat{y}|$. Another interesting observation is that XGB outperforms most outcome models for several datasets (e.g., Boston housing, power plant, airfoil noise, concrete compression strength, naval propulsion), but the underlying decision trees behind XGB might be sensitive to a slight change to the input $\mathbf{x}$, which in turn gets translated into an inaccurate estimation of MRIP and a low correlation between $\widetilde{\mathcal{R}}(\mathbf{x})$ and $|y-\widehat{y}|$.

The MRIP is indicative of the reliability of ML model prediction across a broad range of datasets. Such information can be further exploited to support decision-making activities in risk-sensitive environments. For example, application-specific threshold values can be established to determine when to abstain from ML model's predictions. Given the threshold value, MRIP can be used to determine when to accept ML model prediction, when to abstain. From this standpoint, the MRIP indicator provides an additional layer of safety in addition to the original ML model prediction to prevent counting on ML model prediction that is deemed unreliable, thus increasing the safety assurance of adopting ML solutions in high-risk settings to some extent. Following the information gain-driven approach described before, we determine the threshold value λ over $\widehat{\mathcal{R}}(\mathbf{x})$ for the eight best-performed models. Table 8 displays the specific value of λ corresponding to each dataset. Among all the datasets, power plant and naval propulsion have the highest threshold values of 0.980 as suggested by the information gain-based approach, followed by the auto MPG and Boston housing. The λ value

[5] We examine the effect of different bin sizes and find out that the effect of bin size on the correlation coefficient of best-performed model is trivial.

Table 7
Correlation-based performance assessment of two-stage ML-based MRIP estimation.

Dataset	Outcome model	MRIP model									Baselines						
		DT	LR	RR	LASSO	NN	RF	SVR	XGB	KNN	VarBag	VarA	VarP	MSE	RE_1	RE_2	RE_3
	DT	0.336	0.385	0.391	0.360	0.356	0.450	0.409	0.455	0.378	0.356	0.214	0.079	0.147			
	LR	0.558	0.478	0.483	0.481	0.502	0.616	0.47	0.621	0.485	0.205	0.361	0.104	0.080			
	RR	0.542	0.471	0.472	0.470	0.473	0.608	0.478	0.642	0.464	0.196	0.364	0.106	0.084			
	LASSO	0.562	0.486	0.49	0.488	0.478	0.628	0.484	0.608	0.473	0.204	0.367	0.101	0.101			
Boston Housing	NN	0.554	0.564	0.569	0.555	0.621	0.662	0.573	0.678	0.456	0.290	0.366	0.067	0.079	0.156	0.280	0.192
	RF	0.288	0.455	0.463	0.349	0.414	0.408	0.488	0.500	0.418	0.344	0.313	0.088	0.100			
	SVR	0.615	0.549	0.538	0.549	0.586	**0.681**	0.457	0.676	0.486	0.275	0.402	0.088	0.101			
	XGB	0.258	0.394	0.410	0.322	0.468	0.470	0.438	0.481	0.484	0.402	0.280	0.079	0.098			
	KNN	0.397	0.280	0.276	0.295	0.284	0.497	0.265	0.526	0.349	0.264	0.430	0.087	0.105			
	DT	0.190	0.182	0.182	0.181	0.201	0.326	0.287	0.294	0.339	0.229	0.346	0.015	0.019			
	LR	0.301	0.108	0.106	0.110	0.262	0.351	0.335	0.345	0.391	0.036	0.334	0.026	0.072			
	RR	0.294	0.108	0.106	0.109	0.261	0.355	0.335	0.345	0.391	0.036	0.334	0.026	0.072			
	LASSO	0.301	0.105	0.103	0.107	0.276	0.350	0.333	0.341	0.392	0.036	0.334	0.026	0.072			
Power Plant	NN	0.256	0.186	0.190	0.188	0.248	0.372	0.339	0.344	**0.420**	0.001	0.391	0.022	0.026			
	RF	0.189	0.202	0.207	0.210	0.213	0.294	0.294	0.276	0.308	0.182	0.381	0.016	0.022			
	SVR	0.221	0.197	0.201	0.201	0.239	0.333	0.306	0.308	0.364	0.084	0.399	0.020	0.025			
	XGB	0.170	0.179	0.186	0.182	0.223	0.269	0.276	0.242	0.302	0.162	0.354	0.019	0.024			
	KNN	0.194	0.156	0.157	0.157	0.190	0.308	0.289	0.274	0.338	0.215	0.400	0.014	0.024			
	DT	0.338	0.268	0.351	0.375	0.427	0.385	0.474	0.395	0.381	0.111	0.391	0.059	0.154			
	LR	0.232	0.259	0.293	0.288	0.312	0.312	0.352	0.315	0.253	0.246	0.403	0.077	0.118			
	RR	0.321	0.271	0.233	0.387	0.285	0.31	0.307	0.3	0.261	0.228	0.413	0.073	0.116			
	LASSO	0.271	0.212	0.269	0.250	0.334	0.351	0.371	0.316	0.290	0.223	0.425	0.076	0.118			
Istanbul stock exchange	NN	0.339	0.274	0.255	0.264	0.319	0.366	0.387	0.348	0.318	0.259	0.443	0.089	0.124	0.204	0.239	0.139
	RF	0.357	0.297	0.270	0.268	0.297	0.434	0.428	**0.480**	0.450	0.113	0.448	0.093	0.114			
	SVR	0.226	0.266	0.22	0.342	0.300	0.330	0.379	0.332	0.322	0.311	0.421	0.073	0.116			
	XGB	0.335	0.225	0.235	0.240	0.376	0.378	0.406	0.372	0.395	0.115	0.466	0.090	0.126			
	KNN	0.303	0.235	0.150	0.237	0.290	0.407	0.477	0.437	0.470	0.062	0.470	0.073	0.105			
	DT	0.147	0.127	0.124	0.124	0.208	0.278	0.260	0.265	0.335	0.743	0.132	0.034	0.028			
	LR	0.607	0.175	0.206	0.194	0.663	0.668	0.601	0.663	0.650	0.057	0.385	0.027	0.045			
	RR	0.584	0.169	0.193	0.185	0.663	0.664	0.591	0.659	0.650	0.030	0.389	0.037	0.040			
	LASSO	0.603	0.163	0.186	0.181	0.675	0.674	0.603	0.666	0.656	0.026	0.388	0.034	0.040			
Parkinson's disease score	NN	0.570	0.409	0.407	0.407	0.651	0.668	0.634	0.659	0.659	0.272	0.457	0.025	0.026			
	RF	0.115	0.107	0.107	0.105	0.258	0.335	0.298	0.305	0.421	0.691	0.304	0.035	0.033			
	SVR	0.679	0.462	0.467	0.463	0.739	0.750	0.718	0.739	**0.755**	0.141	0.489	0.040	0.043			
	XGB	0.100	0.113	0.114	0.114	0.243	0.344	0.323	0.305	0.452	0.542	0.244	0.028	0.030			
	KNN	0.342	0.271	0.274	0.272	0.459	0.467	0.448	0.448	0.510	0.572	0.495	0.037	0.024			
	DT	0.329	0.304	0.320	0.330	0.327	0.507	0.396	0.498	0.467	0.329	0.187	0.082	0.047			
	LR	0.459	0.530	0.531	0.533	0.565	0.636	0.605	0.629	0.623	0.303	0.279	0.083	0.056			
	RR	0.523	0.507	0.513	0.508	0.554	**0.648**	0.620	0.626	0.618	0.290	0.281	0.082	0.055			
	LASSO	0.483	0.500	0.506	0.503	0.592	0.643	0.597	0.626	0.618	0.302	0.279	0.083	0.055			
Concrete compression strength	NN	0.370	0.425	0.429	0.420	0.461	0.533	0.492	0.524	0.479	0.298	0.256	0.089	0.102	0.319	0.348	0.152
	RF	0.331	0.387	0.378	0.421	0.431	0.511	0.472	0.517	0.450	0.316	0.213	0.059	0.033			
	SVR	0.344	0.405	0.400	0.388	0.375	0.468	0.429	0.482	0.402	0.360	0.258	0.032	0.081			
	XGB	0.271	0.395	0.361	0.315	0.370	0.531	0.389	0.519	0.413	0.360	0.189	0.070	0.039			
	KNN	0.364	0.475	0.475	0.475	0.476	0.592	0.498	0.604	0.527	0.192	0.355	0.063	0.058			
	DT	0.460	0.368	0.377	0.368	0.493	0.595	0.499	0.568	0.591	0.356	0.190	0.040	0.066			
	LR	0.565	0.410	0.423	0.421	0.595	0.611	0.571	0.598	0.588	0.205	0.336	0.041	0.045			
	RR	0.552	0.428	0.443	0.442	0.603	0.603	0.55	0.588	0.576	0.196	0.337	0.040	0.043			
Airfoil noise	LASSO	0.564	0.428	0.441	0.440	0.624	0.617	0.571	0.598	0.587	0.204	0.336	0.041	0.045	0.070	0.095	0.042
	NN	0.563	0.440	0.448	0.440	0.626	0.639	0.587	0.616	0.618	0.290	0.215	0.058	0.048			
	RF	0.530	0.472	0.484	0.471	0.562	0.619	0.578	0.607	0.640	0.344	0.264	0.031	0.045			
	SVR	0.598	0.574	0.573	0.574	0.662	0.661	0.665	0.664	0.660	0.275	0.243	0.062	0.053			

(*continued on next page*)

Table 7 (*continued*)

Dataset	Outcome model	MRIP model									Baselines						
		DT	LR	RR	LASSO	NN	RF	SVR	XGB	KNN	VarBag	VarA	VarP	MSE	RE_1	RE_2	RE_3
	XGB	0.579	0.447	0.470	0.448	0.576	0.651	0.581	0.647	**0.685**	0.402	0.193	0.049	0.024			
	KNN	0.448	0.415	0.427	0.417	0.543	0.624	0.517	0.608	0.633	0.264	0.236	0.053	0.058			
	DT	0.408	0.407	0.407	0.427	0.425	0.408	0.429	0.406	0.404	0.621	0.359	0.024	0.024			
	LR	0.694	0.702	**0.702**	0.690	0.701	0.695	0.697	0.695	0.694	0.525	0.143	0.027	0.029			
	RR	0.673	0.675	0.675	0.660	0.677	0.674	0.676	0.675	0.673	0.158	0.143	0.027	0.031			
	LASSO	0.655	0.625	0.627	0.605	0.656	0.655	0.651	0.656	0.656	0.303	0.111	0.028	0.148			
Naval propulsion	NN	NaN[a]	NaN	NaN	NaN	0.389	NaN	NaN	0.049	NaN	0.506	0.062	0.021	0.023			
	RF	0.619	0.636	0.635	0.650	0.627	0.618	0.628	0.619	0.618	0.639	0.461	0.027	0.017			
	SVR	NaN	NaN	NaN	NaN	0.326	NaN	NaN	0.171	NaN	0.700	0.435	0.034	0.108			
	XGB	0.583	0.614	0.614	0.662	0.605	0.589	0.597	0.586	0.585	0.594	0.341	0.018	0.031			
	KNN	0.626	0.657	0.658	0.670	0.648	0.632	0.65	0.632	0.628	0.525	0.485	0.016	0.028			
	DT	0.257	0.361	0.356	0.344	0.313	0.443	0.323	0.415	0.432	0.337	0.346	0.141	0.123			
	LR	0.227	0.292	0.311	0.21	0.476	**0.562**	0.529	0.556	0.534	0.254	0.365	0.093	0.071			
	RR	0.298	0.295	0.276	0.235	0.487	0.503	0.462	0.509	0.507	0.248	0.386	0.081	0.068			
	LASSO	0.36	0.308	0.345	0.359	0.403	0.483	0.532	0.553	0.500	0.242	0.393	0.085	0.054			
Auto MPG	NN	0.309	0.440	0.477	0.406	0.394	0.463	0.456	0.444	0.480	0.337	0.465	0.098	0.078	0.207	0.254	0.230
	RF	0.210	0.320	0.336	0.252	0.393	0.458	0.365	0.450	0.426	0.352	0.451	0.110	0.142			
	SVR	0.402	0.463	0.494	0.499	0.468	0.447	0.371	0.420	0.464	0.377	0.478	0.105	0.103			
	XGB	0.233	0.329	0.378	0.274	0.369	0.453	0.395	0.408	0.422	0.339	0.454	0.091	0.100			
	KNN	0.292	0.455	0.471	0.484	0.419	0.459	0.421	0.447	0.421	0.301	0.487	0.096	0.126			

[a] As the second ML model generates an identical value for MRIP, the correlation between the prediction error and the MRIP value becomes invalid.

of Parkinson's disease score is the lowest among all the datasets.

Next, we further look into the performance of the proposed ML-based framework in detecting the instances violating the precision requirement that is preset for y. To be specific, we treat instances with $|y - \widehat{y}|$ larger than ε as violated instances. Next, we fix λ's value for each dataset and count the number of instances with $\widehat{\mathcal{R}}(\mathbf{x})$ larger than λ. Our objective is to identify the proportion of violated cases that are correctly identified based on the threshold parameter λ. In addition, we are also interested in knowing the proportion of violated instances correctly detected based on the threshold parameter λ and the value of ε. Note that ideally, when ε changes, the MRIP model and relevant parameters, such as λ, should be derived from scratch to reflect the change of ε. In this example, our goal is to use the MRIP model trained with a fixed value of ε and δ as an approximation to reflect the capability of MRIP in detecting high-risk instances, where we assume the risk of an instance is positively correlated with the prediction error.

Fig. 11 illustrates the variation of the percentage of violated instances that are detected based on the threshold parameter λ when ε gradually increases from the predetermined value that is set according to Eq. (11) to the maximum value of $|y - \widehat{y}|$ over the test data. As can be seen in Fig. 11, λ acts as an effective safety guard when ε gradually increases. In particular, except the case of auto MPG, the proportion of violated instances detected by λ rises significantly although with some slight fluctuations in some cases. Considering that the larger value of ε typically indicates a higher level of risk, the threshold-based detection built on $\widehat{\mathcal{R}}(\mathbf{x})$ exhibits an appealing feature as it demonstrates a better detection capability for cases with higher risk. From Fig. 11, we also observe different degrees of fluctuations occur in the examples of power plant, auto MPG, and Parkinson's. In particular, the fluctuation associated with the auto MPG is the strongest among all fluctuations. From Table 5, it can be seen that auto MPG has the smallest size of observations among all the datasets. The limited number of training set might lead to a poor performance of the ML model trained for outcome prediction, which could partially explain the severe fluctuations in the case of auto MPG. In addition, other factors, such as the complexity of the learning problem, the data sparseness, could also have an essential impact on the performance of the trained ML model. For example, if the observations are scare around a specific region of $\boldsymbol{x}$, then the estimation of the MRIP indicator might not be able to accurately reflect the true reliability of the individual model prediction for that particular region. The inaccurate reliability estimation could get propagated into the ML model, thus leading to the fluctuations as seen in Fig. 11.

In addition, with respect to the best-performing model, we also examine its false positive rate, that is the proportion of data with the prediction error $|y - \widehat{y}|$ larger than the predefined threshold ε defined in Eq. (11), but the proposed approach generates a reliability estimate higher than the λ value reported in Table 8. From Table 9, it can be observed that the developed model has a low false positive rate for most datasets, only two datasets Istanbul stock exchange and Parkinson's disease score have a relatively high false positive rate. A further examination finds out that the scarcity in the Parkinson's disease score results in a large average prediction error, which is nearly 38% of the value of ε; while in the case of Istanbul stock exchange, the data scarcity results in a high overlap in the search of neighbors when deriving the MRIP value, thus leading to a low level of distinction in the MRIP value and a high false positive rate.

5. Discussion

The proposed reliability metric has important practical implications along the development of decision support systems for managing ML models in critical applications with serious safety and reliability concerns. In the first place, the developed MRIP indicator provides a quantitative and individualized indicator to support important decisions in determining when to count on ML model prediction, when to request

Table 8
The threshold value λ decided by the information gain-based approach.

	Boston Housing	Power plant	Istanbul stock exchange	Parkinson's disease score	Airfoil noise	Concrete compression strength	Naval propulsion	Auto MPG
λ	0.838	0.980	0.596	0.273	0.616	0.596	0.980	0.919

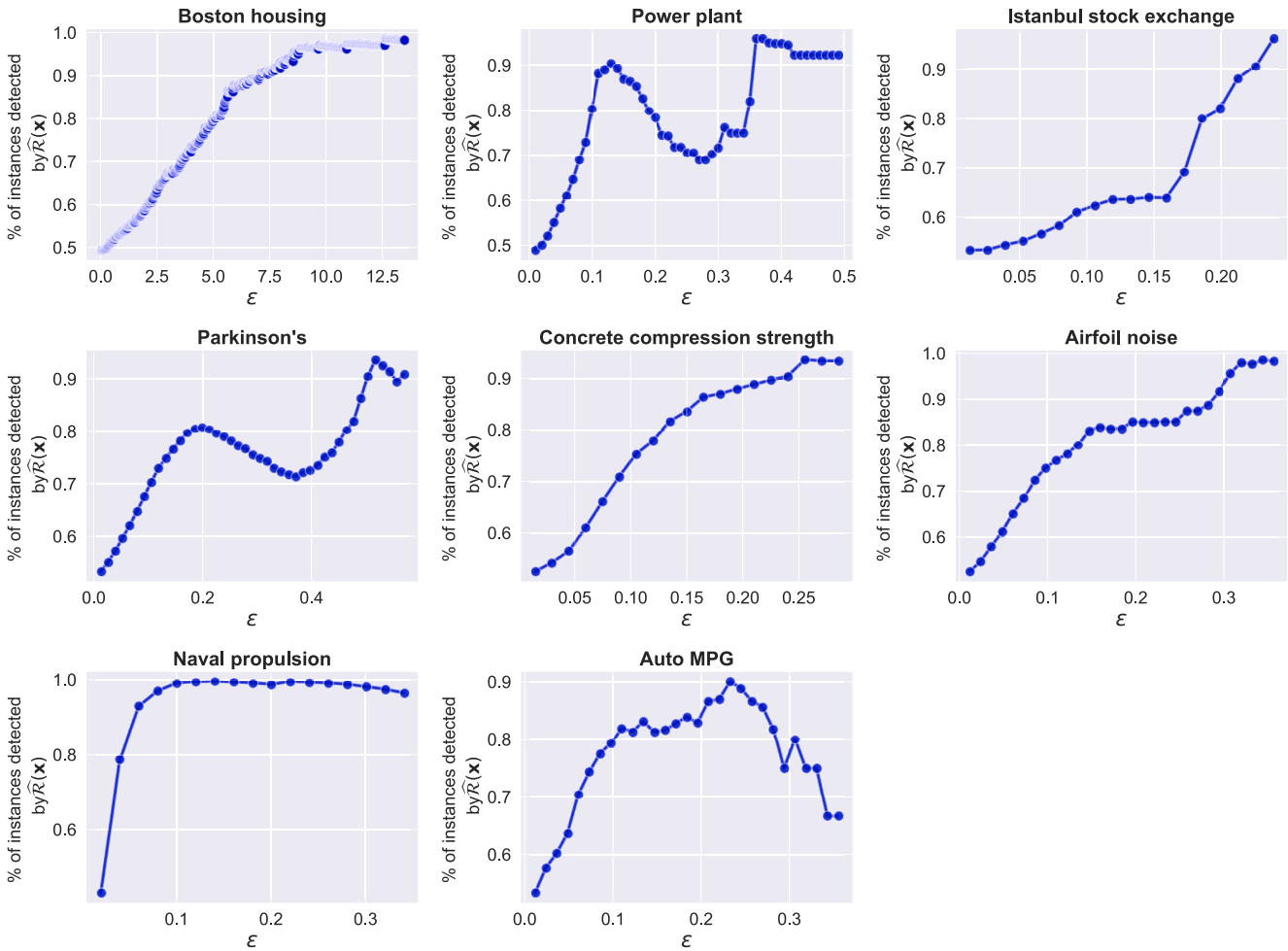


Fig. 11. Variation of the proportion of violated instances detected based on λ for the best performed combination of outcome model and MRIP model when ε gradually increases.

Table 9
False positive rate of the best-performing model with respect to each dataset.

Dataset	Boston Housing	Power plant	Istanbul stock exchange	Parkinson's disease score	Airfoil noise	Concrete compression strength	Naval propulsion	Auto MPG
False positive rate	0.110	0.004	0.452	0.381	0.201	0.213	0.0006	0.064

the intervention of human experts. The development of such an indicator thus substantially facilitates proactive risk management of deployed ML models in high-stakes environment on a case-by-case basis. Secondly, as illustrated in the computational study, the two-stage ML framework for MRIP estimation is more sensitive to cases with a larger prediction error. Such feature enables to detect adverse events with a probability approximately proportionate to the prediction error. In other words, adverse events with more severe outcome have a higher chance to be detected by the developed reliability metric. From this perspective, the developed approach has a potential to diminish unintended outcomes of ML models in risk-sensitive applications significantly. Thirdly, the developed indicator offers a means to fully incorporate heterogeneous noise associated with both $\mathbf{x}$ and y arising from a rich set of sources in the reliability indicator. With the risk arising from input data and observation quantified appropriately, we make ML system aware of its limitations in the decision environment, such that it can recognize and inform end users when the overall risk is above a threshold and the system is no longer able to make reliable predictions or decisions. The decision support as empowered by the developed approach contributes to the safe and reliable adoption of ML in high-stakes decision-making environments.

Although the proposed approach shows a great promise in estimating model reliability for individual prediction, a rigorous approach is still lacking for guaranteeing the soundness of the estimated MRIP as the

scarcity of neighbors might lead to unreliable MRIP estimate. In this paper, we follow a rigorous procedure to examine the pairwise correlations between $|y-\widehat{y}|$ and $\widetilde{\mathcal{R}}(\mathbf{x})$ across all the considered models as reported in Table 7. To detect model performance issues, we show how to compute false positive rates (see Table 9) and show how to detect instances that violate the precision requirements (i.e., $|y-\widehat{y}|>\varepsilon$) (see Fig. 11). To play safe in practice, sanity checks can be quickly performed to diagnose whether the model is failing. For example, the increase of ε should give rise to an elevated detection rate of cases violating the predetermined precision requirement; if the model exhibits a downward trend with the increase of ε, then it is probably insufficient to be used for MRIP estimation. Another measure for model diagnosis is to scrutinize the false positive rate committed by the developed model. A high false positive rate is an informative indicator on the failure of the model in MRIP estimation as the model is unable to correctly differentiate prediction failure vs prediction success. The failure of the model is in part due to the scarcity in searching for enough number of neighbors around $\mathbf{x}$ or scarce training data for sufficiently training the stage-one ML model. From the computational perspective, the two suggested model diagnosis measures will not incur severe computational overhead as they only involve model inference.

6. Conclusion

The reliability of the ML model regarding its prediction on each individual input has surfaced as an important consideration when adopting ML models in risk-sensitive applications. In this paper, we propose an objective general-purpose probabilistic indicator $\mathcal{R}(\mathbf{x})$ to measure the ML model reliability for individual prediction, where $\mathcal{R}(\mathbf{x})$ is defined as the probability of the observed difference between the ML model prediction and the actual observation falling within a small interval while the input $\mathbf{x}$ varies within a small range subject to a distance constraint, namely $\mathcal{R}(\mathbf{x})=P(|\widehat{y}^*-y^*|\leq\varepsilon\,|\mathbf{x}^*\in B(\mathbf{x}))$. The proposed reliability indicator $\mathcal{R}(\mathbf{x})$ extensively exploits the local information associated with the density of the input $\mathbf{x}$ and the ML model fitness, and it follows two fundamental principles when judging whether a ML model prediction is reliable: the density criteria and the local criteria. Hence, it has the advantage of characterizing the inherent aleatory and epistemic uncertainty of heterogeneous nature arising from both $\mathbf{x}$ and the ML model. In this study, we also formulate a two-stage ML-based approach to estimate the MRIP, where state-of-the-art ML algorithms are explored to connect the input feature $\mathbf{x}$ and the corresponding MRIP value. An information gain-based method is developed to help determine the threshold value of $\mathcal{R}(\mathbf{x})$ so as to determine when to decline ML predictions. Comprehensive computational studies are performed to demonstrate the performance of the developed MRIP metric in reflecting the reliability of ML model for individual predictions. A rich set of ML models trained with eight UCI datasets are used to showcase the robust performance of the proposed approach for MRIP modeling and estimation.

The following research directions need to be investigated further in the future. In this paper, we only examine the performance of the MRIP indicator in handling the heterogeneous noise from the input and the ML model. In the future, it will be interesting to study the behaviour of the MRIP indicator when dealing with other forms of risks, such as, model form inadequacy, adversarial attacks/perturbation of model inputs. In addition, we only showcase the application of MRIP in numerical prediction problems. It is worth investigating the adoption of the proposed MRIP indicator in other contexts, such as classification problem, autonomous driving. Secondly, the MRIP problem can be formulated in other ways. For example, the learning problem can also be framed as $\mathcal{R}(\mathbf{x})=P(|\widehat{y}^*-y^*|\leq\varepsilon\,|\mathbf{x}^*\in B(\mathbf{x}))\geq\alpha$. By specifying the minimum level of confidence that is required, ML models can be constructed to learn the minimum value of ε for any fixed $B(\mathbf{x})$. Or even in an extreme case, by setting α's value as 1, we can train a ML model to learn a minimum bound that contains the actual value of y given the ML model prediction $\widehat{y}$. Thirdly, we assume that ε and δ in the reliability estimator take constant values for the sake of consistency in the demonstration of experimental comparisons. If prior knowledge on the variation of ε and δ in accordance with $\mathbf{x}$ is available, then ε and δ can take varying values dependent on $\mathbf{x}$. Fourthly, the effect of several essential factors on the estimation of MRIP indicator, such as the size of training set, dataset sparseness, and distribution of the data, needs to be rigorously studied for different critical applications (e.g., cancer diagnosis) in a future study. Last but not least, the reliability estimation pertaining to a specific ML model prediction is established upon another ML approach, which creates an inevitable recursive situation for reliability estimation. Although computational results and quantitative comparisons suggest that the impact of recursive estimation of MRIP is not large given the benefits that MRIP provides us, it does not mean that the ML model at the second stage is theoretically guaranteed to have better performance than the ML model in the first stage. In the future, it will be meaningful to investigate and develop a non-recursive approach for estimation of individual prediction reliability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Xiaoge Zhang is an Assistant Professor in the Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University (PolyU). He serves as Associate Director of Behaviour and Knowledge Engineering Research Centre (BAKE) at PolyU. He was a recipient of the Chinese Government Award for Outstanding Self-financed Students Abroad in 2017. He has published more than 70 research articles in leading academic journals, such as Nature Communications, Risk Analysis, Decision Support Systems, IEEE Transactions on Information Forensics and Security, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Cybernetics, IEEE Transactions on Reliability, IEEE Transactions on Intelligent Transportation Systems, Knowledge-Based Systems, and Annals of Operations Research, among others. His current research interests include risk analysis and reliability assessment of artificial intelligence and machine learning algorithms in high-stakes decision settings. He is on the editorial board of Journal of Organizational Computing and Electronic Commerce. He is a member of INFORMS, IEEE and IISE.

Indranil Bose is Distinguished Professor of Management Information Systems at the NEOMA Business School. He holds a BTech from the Indian Institute of Technology, MS from the University of Iowa, and MS and PhD from Purdue University. His research interests are in business analytics, digital transformation, information security, and management of emerging technologies. His publications have appeared in MIS Quarterly, Journal of the MIS, Communications of the ACM, Communications of the AIS, Computers and Operations Research, Decision Support Systems, Electronic Markets, European Journal of Operational Research, Information & Management, International Journal of Production Economics, Journal of Organizational Computing and Electronic Commerce, Journal of the American Society for Information Science and Technology, Operations Research Letters, Technological Forecasting and Social Change, etc. He serves as Senior Editor of Decision Support Systems, Pacific Asia Journal of the AIS, and Journal of Organizational Computing and Electronic Commerce; Associate Editor of Communications of AIS; and Editorial Review Board member of Information Systems Research.